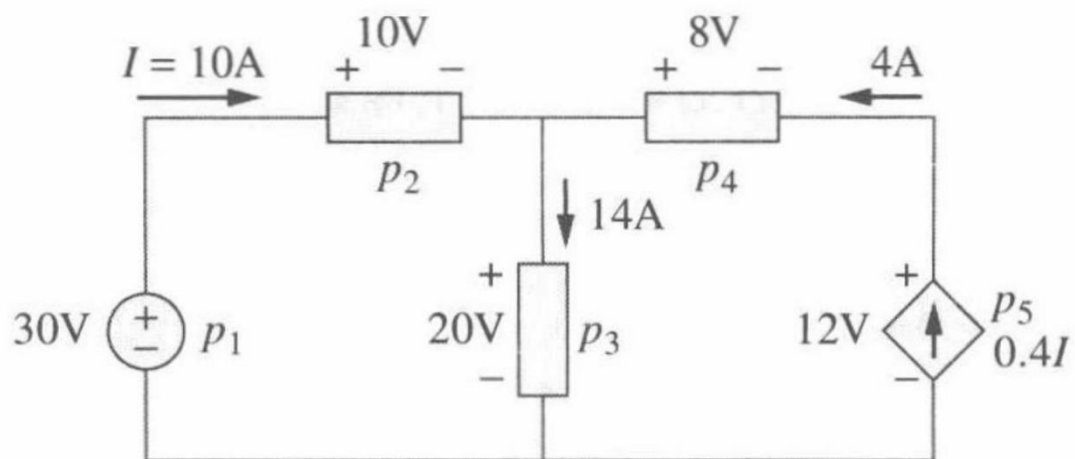


1.18 计算下图中各个元件的吸收的功率。



解：

$$p_1 = 30(-10) = -300 \text{ W}$$

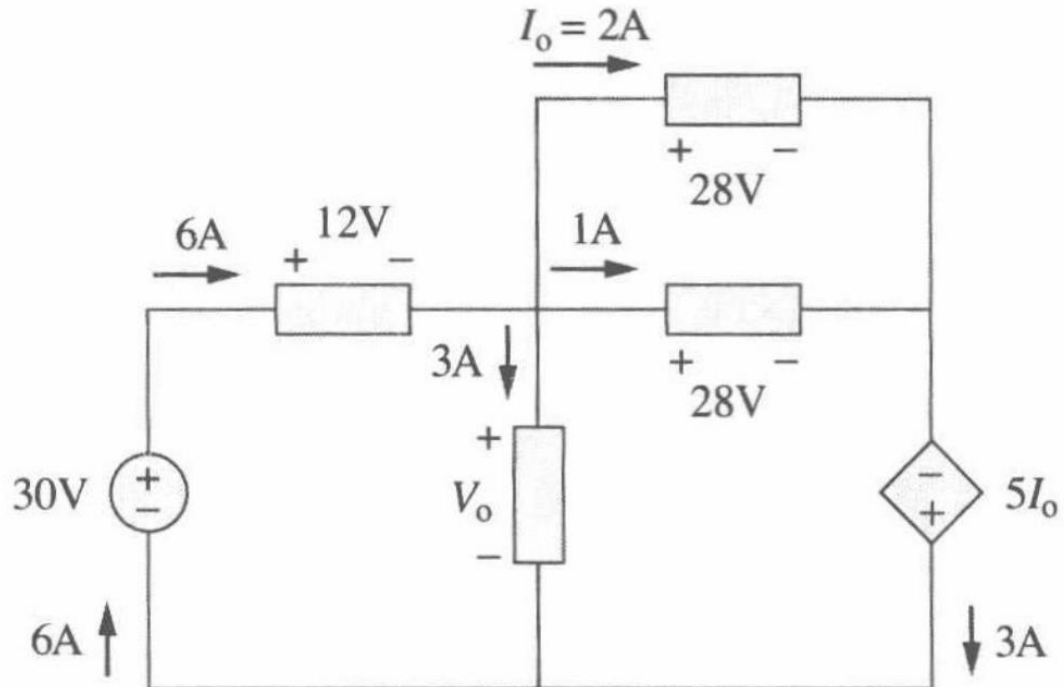
$$p_2 = 10(10) = 100 \text{ W}$$

$$p_3 = 20(14) = 280 \text{ W}$$

$$p_4 = 8(-4) = -32 \text{ W}$$

$$p_5 = 12(-4) = -48 \text{ W}$$

1.20 计算下图中的 V_o 以及每个元件吸收的功率。



解：

$$30\text{V 电压源 } p = 30 \times (-6) = -180 \text{ W}$$

$$12\text{V 元件 } p = 12 \times 6 = 72 \text{ W}$$

$$2\text{A 电流流过的 } 28\text{V 元件 } p = 28 \times 2 = 56 \text{ W}$$

1A 电流流过的 28V 元件 $p = 28 \times 1 = 28 \text{ W}$

5I_o 受控源 $p = 5 \times 2 \times (-3) = -30 \text{ W}$

因为电路中所有元件吸收的功率和为零

$3V_o = 180 - 72 - 56 - 28 + 30 = 54 \text{ W}$

$V_o = 18 \text{ V}$.

1.23 一台 1.8kW 的热水器需要 15min 烧开一定量的水，如果一天烧一次水，并且电费为 10 美分/kW·h，计算工作 30 天需要的电费。

解：

$W = pt = 1.8 \times (15/60) \times 30 \text{ kWh} = 13.5 \text{ kWh}$

$C = 10 \times 13.5 = 135$

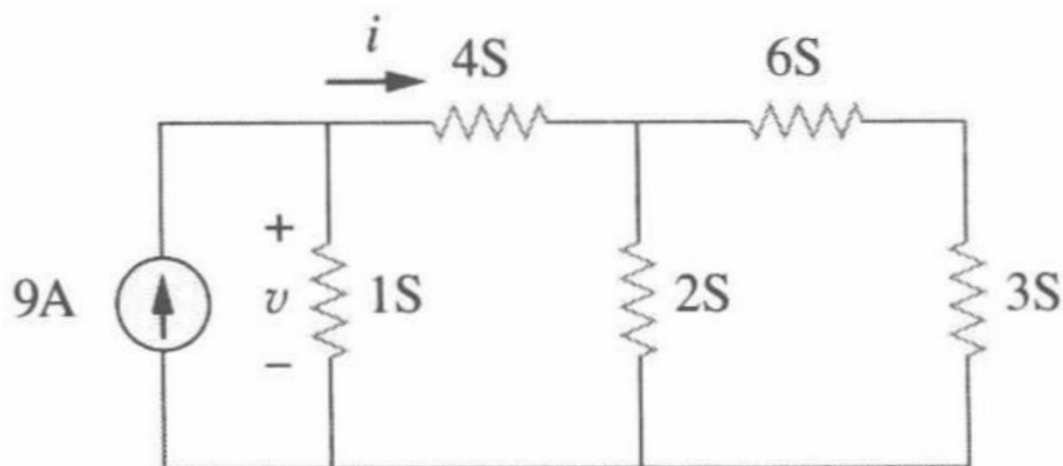
1.36 电池的额定容量可以用 A·h 或者 W·h 表示。A·h 通过 W·h 除以标准的 12V 获得。如果一块汽车电池的额定容量为 20A·h：(a) 求该电池工作 15 分钟所能提供的最大电流；(b) 如果该电池以 2mA 的电流放电，则能持续放电多少天？

解：

(a) $I = 20/0.25 = 80 \text{ A}$

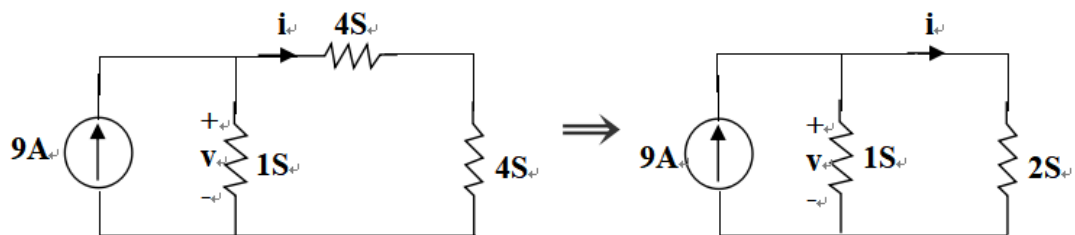
(b) $(20/0.002)/24 = 416.7$

2.33 求下图所示电路的 v 和 i



解：

把电路进行化简：

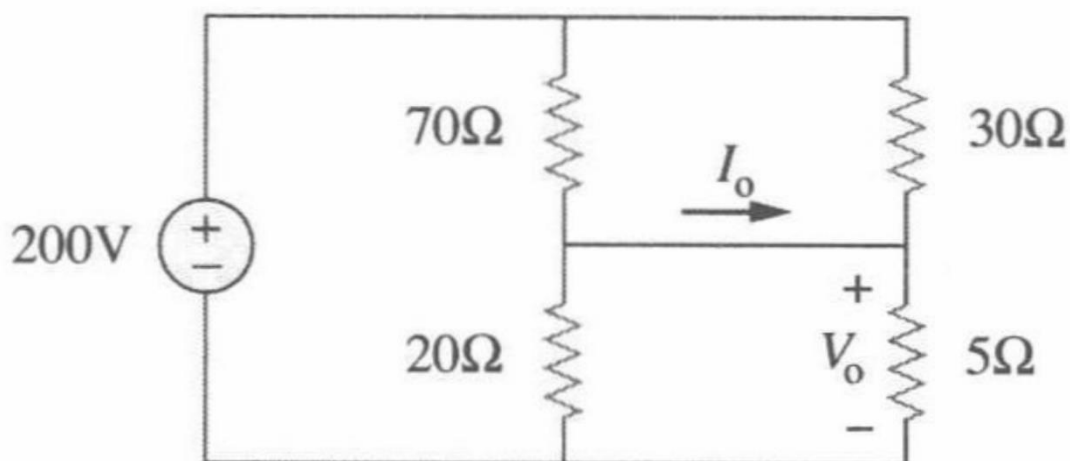


$$6S \parallel 3S = \frac{6 \times 3}{9} = 2S \quad \text{and} \quad 2S + 2S = 4S$$

运用分流定理:

$$i = \frac{1}{1 + \frac{1}{2}}(9) = 6A, \quad v = 3(1) = 3V$$

2.35 计算下图所示电路的 V_o 和 I_o



解:

把电路进行化简:

$$70 \parallel 30 = \frac{70 \times 30}{100} = 21\Omega, \quad 20 \parallel 5 = \frac{20 \times 5}{25} = 4\Omega$$

$$i = \frac{200}{21 + 4} = 8A$$

$$v_1 = 21i = 168V, \quad v_o = 4i = 32V$$

$$i_1 = \frac{v_1}{70} = 2.4A, \quad i_2 = \frac{v_o}{20} = 1.6A$$

根据 KCL:

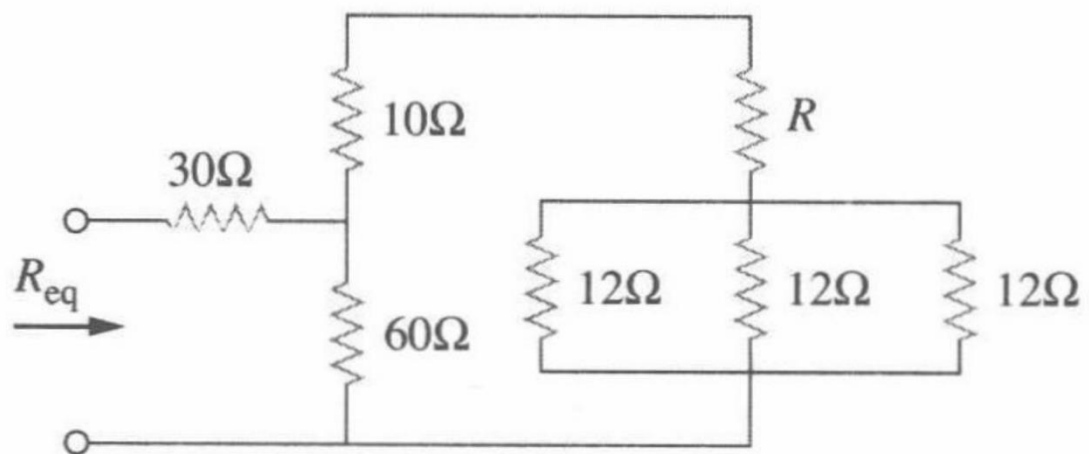
$$i_1 = i_2 + I_o$$

$$2.4 = 1.6 + I_o$$

$$I_o = 0.8A$$

所以 $v_o = 32 \text{ V}$, $i_o = 800 \text{ mA}$

2.41 如果下图所示电路中的 $R_{eq}=50$ ，求 R



解：

设 R_0 为三个并联 12Ω 电阻的等效电阻

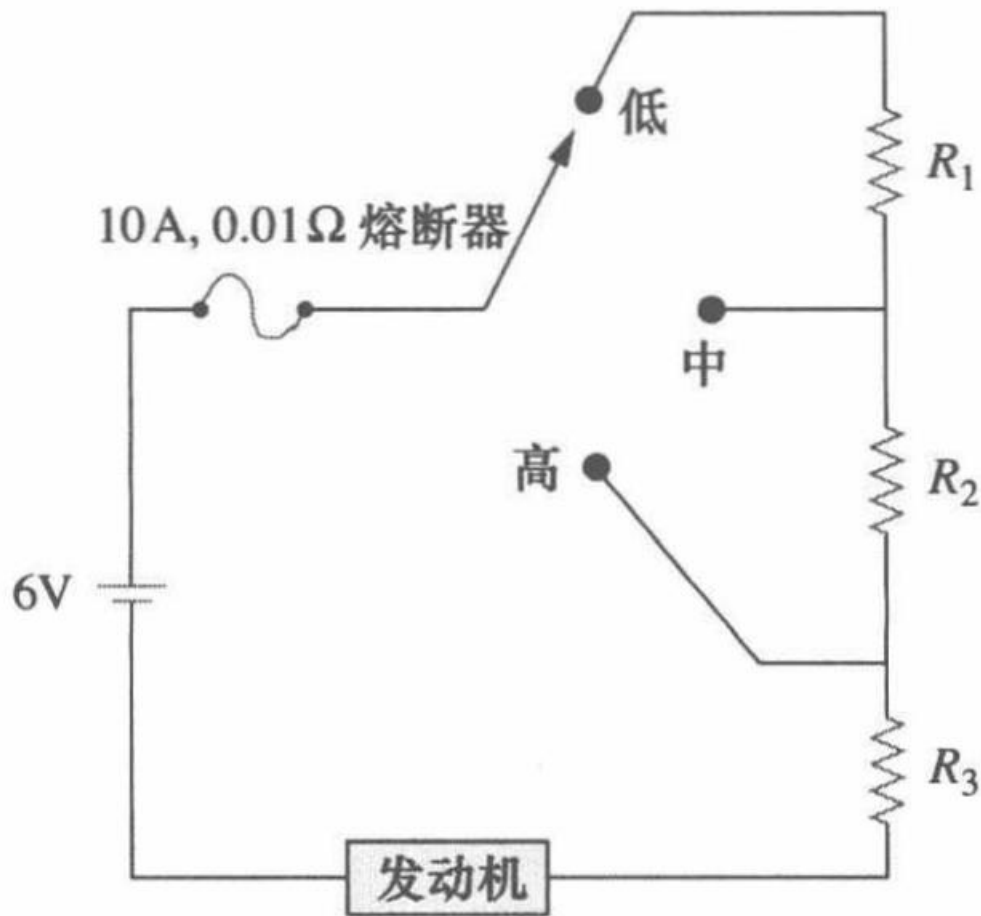
$$\frac{1}{R_0} = \frac{1}{12} + \frac{1}{12} + \frac{1}{12} \longrightarrow R_0 = 4$$

$$R_{eq} = 30 + 60 \parallel (10 + R_0 + R) = 30 + 60 \parallel (14 + R)$$

$$50 = 30 + \frac{60(14 + R)}{74 + R} \longrightarrow 74 + R = 42 + 3R$$

$$R = 16 \Omega$$

2.74 下图所示电路用于控制发动机的转速，当开关掷于高、中、低三个不同位置时，发动机电流分别为 5A、3A 和 1A，可以用一个 $20\text{m}\Omega$ 的负载电阻作为该发动机的电路模型，求串联降压电阻 R_1 、 R_2 、 R_3



解：

开关在上面时

$$6 = (0.01 + R_3 + 0.02) \times 5 \quad R_3 = 1.17 \Omega$$

开关在中间时

$$6 = (0.01 + R_2 + R_3 + 0.02) \times 3 \quad R_2 + R_3 = 1.97$$

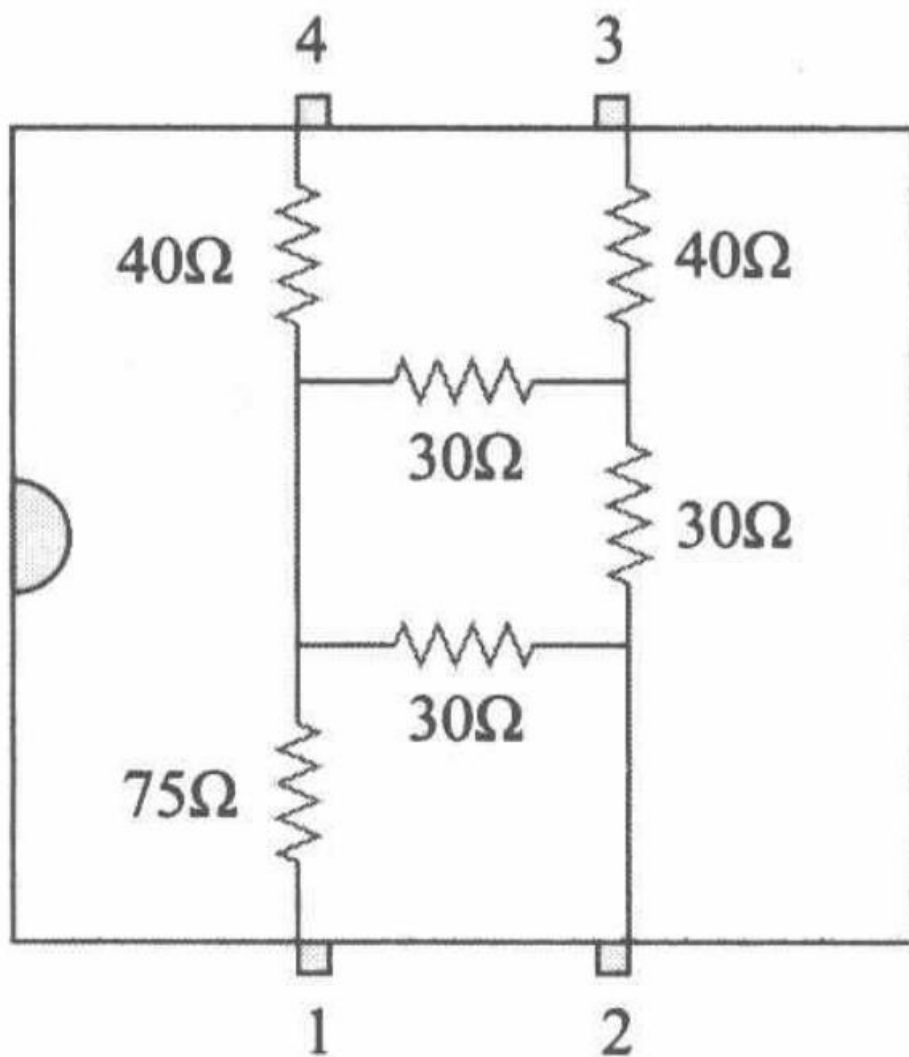
$$R_2 = 1.97 - 1.17 = 0.8 \Omega$$

开关在下面时

$$6 = (0.01 + R_1 + R_2 + R_3 + 0.02) \times 1 \quad R_1 + R_2 + R_3 = 5.97$$

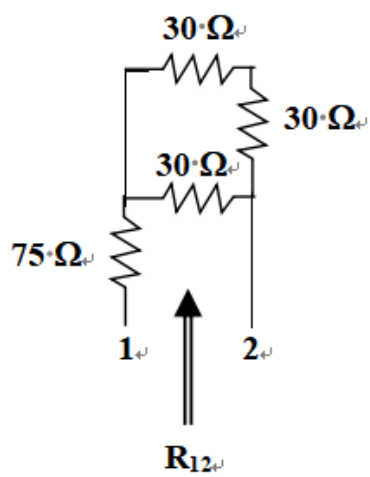
$$R_1 = 5.97 - 1.97 = 4 \Omega$$

2.82 某电阻列阵的引脚图如下图所示，求下述引脚之间的等效电阻：(a) 1 与 2；(b) 1 与 3；(b) 1 与 4。



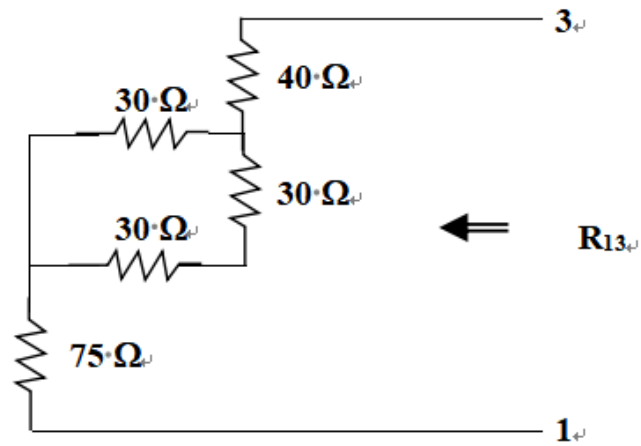
解：

(a)



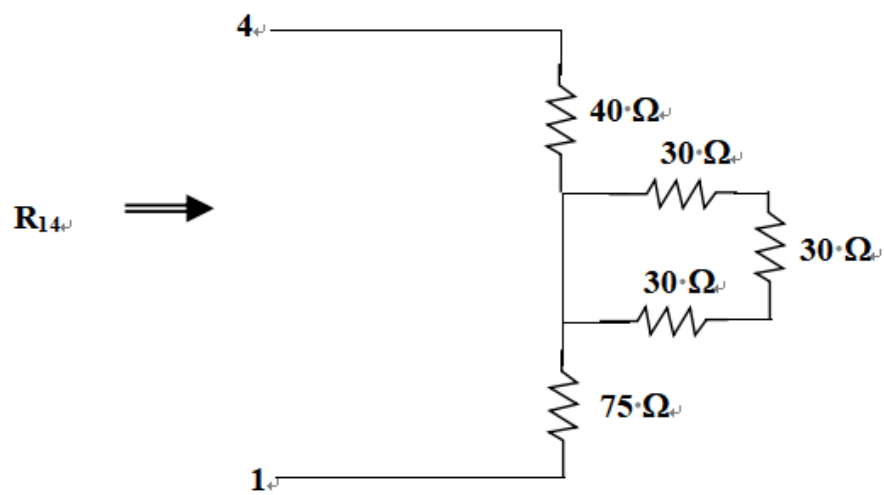
$$R_{12} = 75 + 30 \times 60 / (30 + 60) = 75 + 20 = \mathbf{95 \, \Omega}$$

(b)



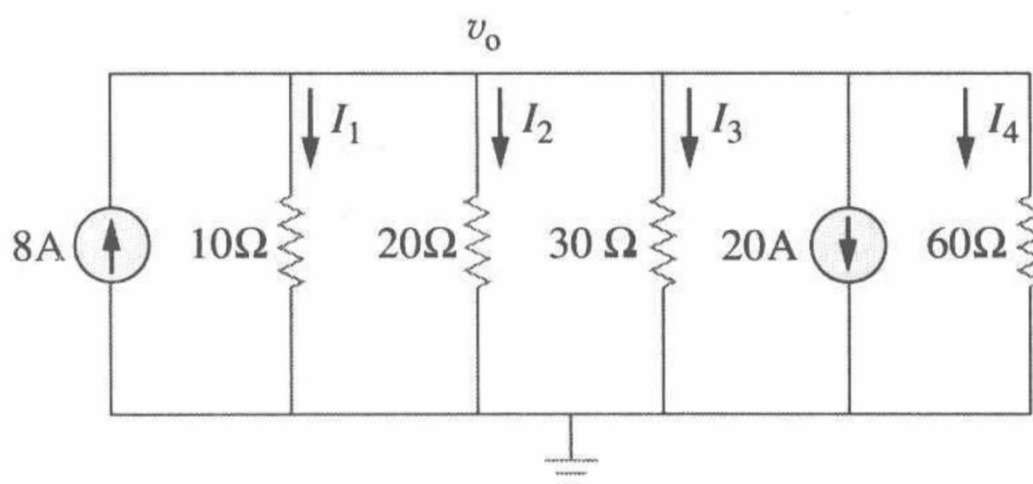
$$R_{13} = 75 + [30 \times 60 / (30 + 60)] + 40 = 135 \, \Omega$$

(c)



$$R_{14} = 40 + 75 = \mathbf{115 \, \Omega}$$

3.3 计算下图所示电路中的电流 $i_1 \sim i_4$ 以及电压 v_o



解:

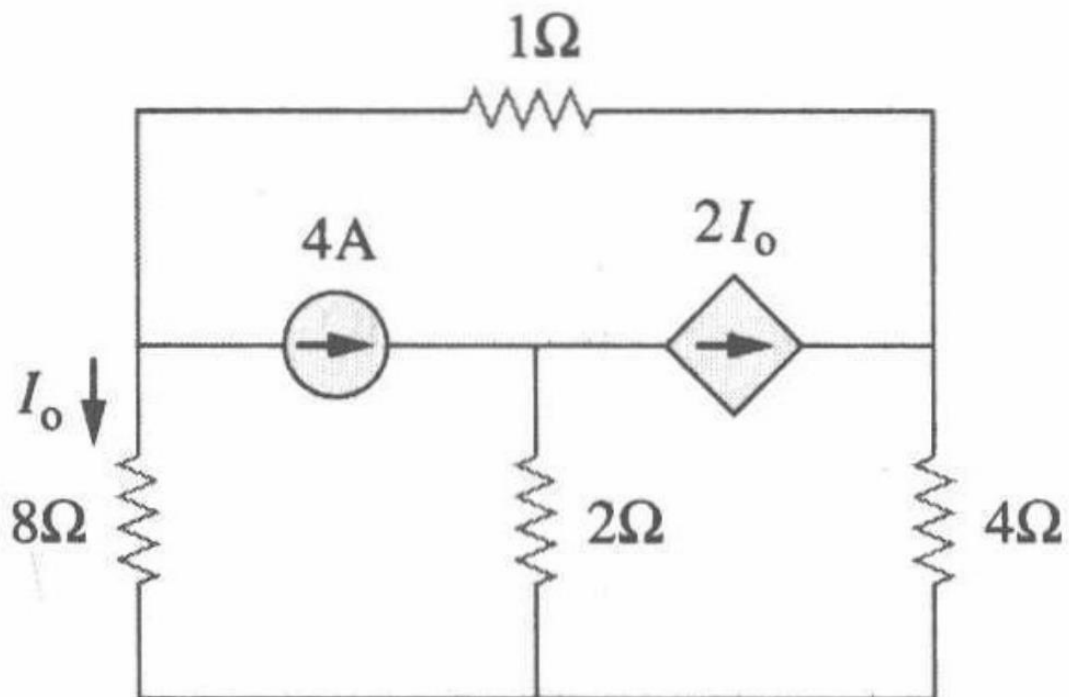
运用 KCL:

$$-8 + \frac{v_o}{10} + \frac{v_o}{20} + \frac{v_o}{30} + 20 + \frac{v_o}{60} = 0 \text{ or } v_o = -60 \text{ V}$$

$$i_1 = \frac{v_o}{10} = -6 \text{ A}, i_2 = \frac{v_o}{20} = -3 \text{ A},$$

$$i_3 = \frac{v_o}{30} = -2 \text{ A}, i_4 = \frac{v_o}{60} = 1 \text{ A}.$$

3.10 计算下图所示电路中的 I_o



解：

在节点 1. $[(v_1-0)/8] + [(v_1-v_3)/1] + 4 = 0$

在节点 2. $-4 + [(v_2-0)/2] + 2i_o = 0$

在节点 3. $-2i_o + [(v_3-0)/4] + [(v_3-v_1)/1] = 0$

可以得到 $i_o = v_1/8$

所以

$$1.125v_1 - v_3 = -4 \quad (1)$$

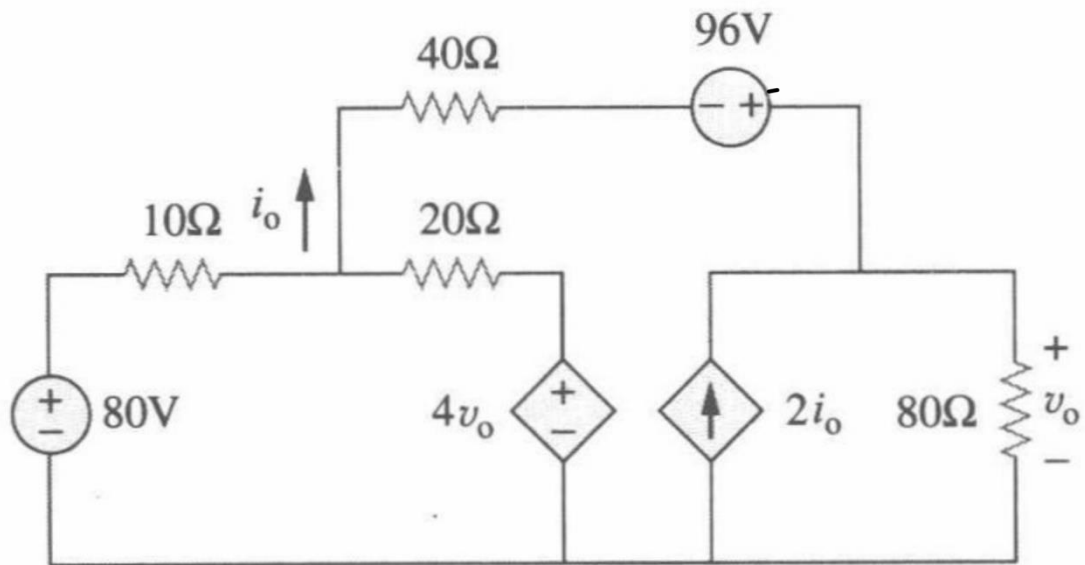
$$0.25v_1 + 0.5v_2 = 4 \quad (2)$$

$$-1.25v_1 + 1.25v_3 = 0 \quad (3)$$

解得：

$$i_o = 32/8 = -4 \text{ amps.}$$

3.30 利用节点分析法计算下图所示电路中的 v_o 和 i_o



解:

在节点 1,

$$\begin{aligned} [(v_1-80)/10]+[(v_1-4v_o)/20]+[(v_1-(v_o-96))/40] &= 0 \text{ or} \\ (0.1+0.05+0.025)v_1 - (0.2+0.025)v_o &= \\ 0.175v_1 - 0.225v_o &= 8-2.4 = 5.6 \end{aligned} \quad (1)$$

在节点 2,

$$\begin{aligned} -2i_o + [((v_o-96)-v_1)/40] + [(v_o-0)/80] &= 0 \text{ and } i_o = [(v_1-(v_o-96))/40] \\ -2[(v_1-(v_o-96))/40] + [((v_o-96)-v_1)/40] + [(v_o-0)/80] &= 0 \\ -3[(v_1-(v_o-96))/40] + [(v_o-0)/80] &= 0 \text{ or} \\ -0.075v_1 + (0.075+0.0125)v_o &= 7.2 = \\ -0.075v_1 + 0.0875v_o &= 7.2 \end{aligned} \quad (2)$$

联立(1) 和 (2) :

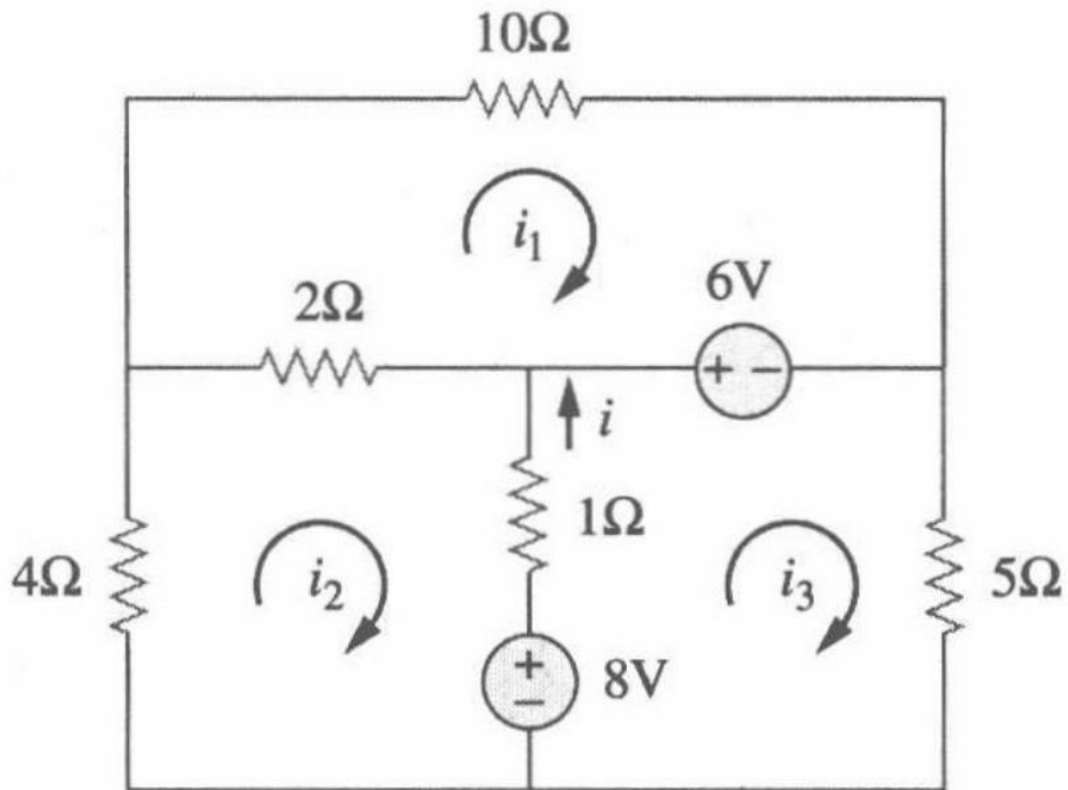
$$\begin{aligned} \begin{bmatrix} 0.175 & -0.225 \\ -0.075 & 0.0875 \end{bmatrix} \begin{bmatrix} v_1 \\ v_o \end{bmatrix} &= \begin{bmatrix} 5.6 \\ 7.2 \end{bmatrix} \text{ or} \\ \begin{bmatrix} v_1 \\ v_o \end{bmatrix} &= \frac{\begin{bmatrix} 0.0875 & 0.225 \\ 0.075 & 0.175 \end{bmatrix}}{0.0153125-0.016875} \begin{bmatrix} 5.6 \\ 7.2 \end{bmatrix} = \frac{\begin{bmatrix} 0.0875 & 0.225 \\ 0.075 & 0.175 \end{bmatrix}}{-0.0015625} \begin{bmatrix} 5.6 \\ 7.2 \end{bmatrix} \end{aligned}$$

$$v_1 = -313.6-1036.8 = -1350.4$$

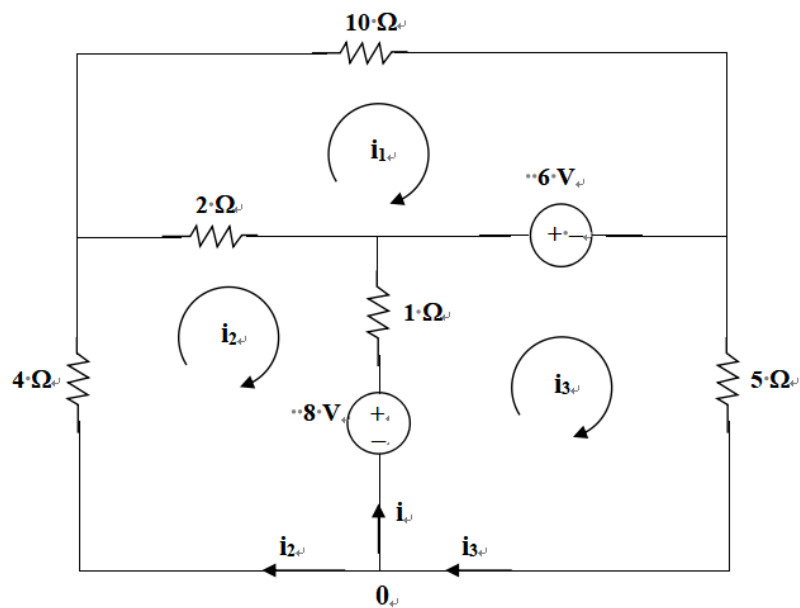
$$v_o = -268.8-806.4 = -1.0752 \text{ kV}$$

$$\text{and } i_o = [(v_1-(v_o-96))/40] = [(-1350.4-(-1075.2-96))/40] = -4.48 \text{ A.}$$

3.41 利用网孔分析法计算下图所示电路中的电流 i



解:



回路 1,

$$6 = 12i_1 - 2i_2 \longrightarrow 3 = 6i_1 - i_2 \quad (1)$$

回路 2,

$$-8 = -2i_1 + 7i_2 - i_3 \quad (2)$$

回路 3,

$$-8 + 6 + 6i_3 - i_2 = 0 \longrightarrow 2 = -i_2 + 6i_3 \quad (3)$$

写成矩阵形式,

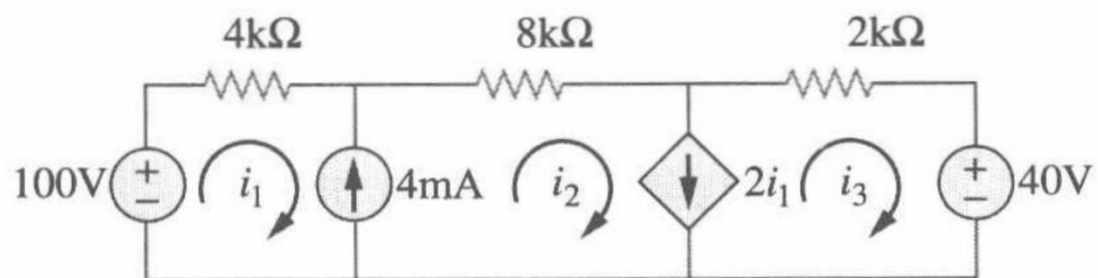
$$\begin{bmatrix} 6 & -1 & 0 \\ 2 & -7 & 1 \\ 0 & -1 & 6 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \\ 2 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 6 & -1 & 0 \\ 2 & -7 & 1 \\ 0 & -1 & 6 \end{vmatrix} = -234, \quad \Delta_2 = \begin{vmatrix} 6 & 3 & 0 \\ 2 & 8 & 1 \\ 0 & 2 & 6 \end{vmatrix} = 240$$

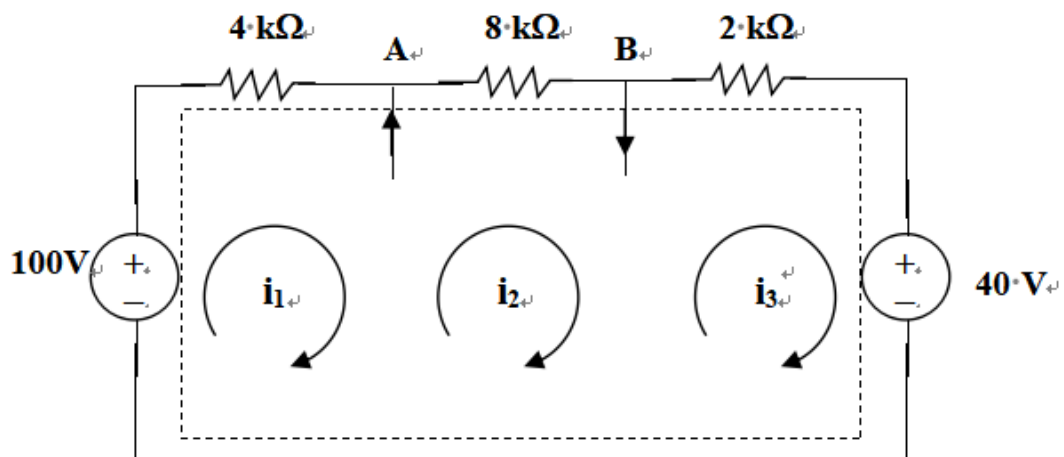
$$\Delta_3 = \begin{vmatrix} 6 & -1 & 3 \\ 2 & -7 & 8 \\ 0 & -1 & 2 \end{vmatrix} = -38$$

$$i = i_3 - i_2 = \frac{\Delta_3 - \Delta_2}{\Delta} = \frac{-38 - 240}{-234} = 1.188 \text{ A}$$

3.62 求下图所示电路中的网孔电流 i_1 、 i_2 和 i_3



解:



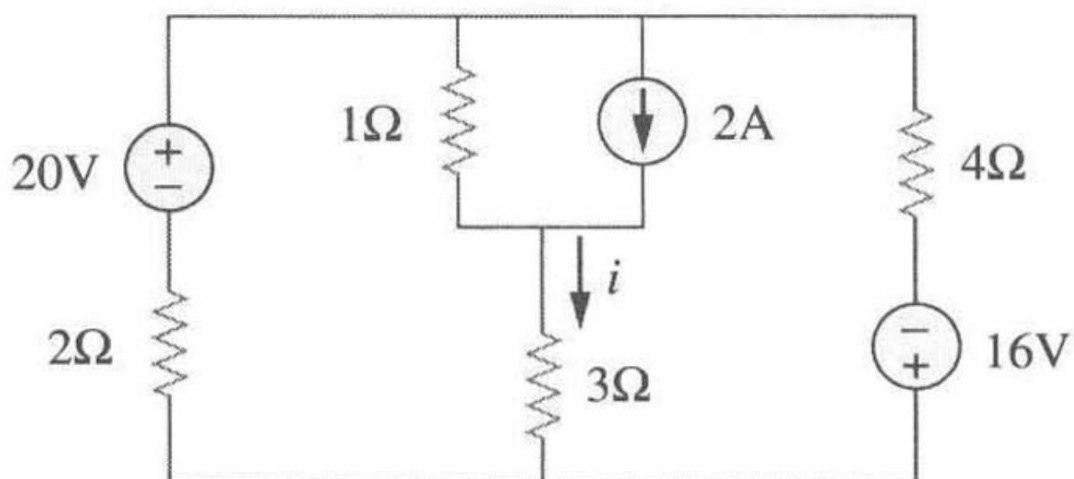
对于超网孔, $-100 + 4i_1 + 8i_2 + 2i_3 + 40 = 0$ or $30 = 2i_1 + 4i_2 + i_3$ (1)

在节点 A, $i_1 + 4 = i_2$ (2)

在节点 B, $i_2 = 2i_1 + i_3$ (3)

解得 $i_1 = 2 \text{ mA}$, $i_2 = 6 \text{ mA}$, $i_3 = 2 \text{ mA}$.

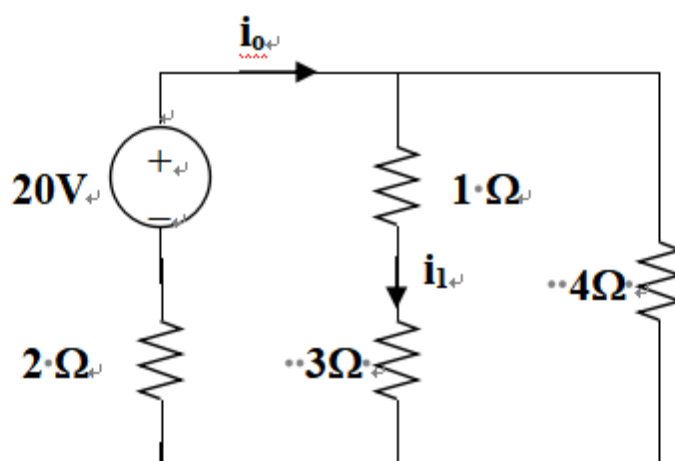
4.15 利用叠加定理确定下图所示电路中的 i , 并计算传递给 3Ω 电阻的功率。



解:

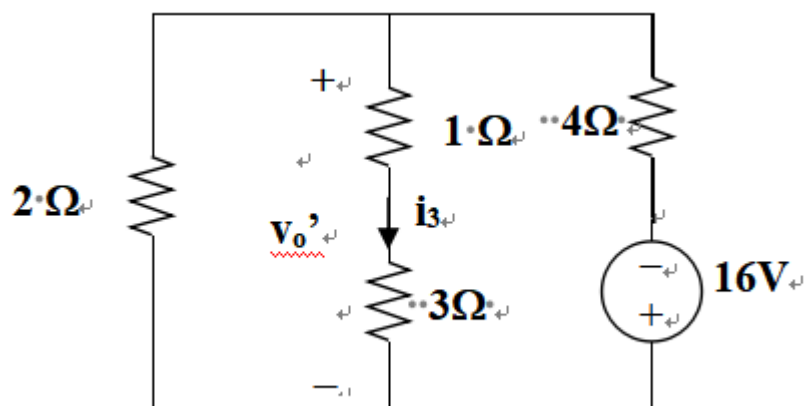
令 $i = i_1 + i_2 + i_3$, i_1, i_2, i_3 是经过 20-V, 2-A, 16-V 电源的电流.

对于 i_1 ,



$$4 \parallel (3 + 1) = 2 \text{ ohms}, \text{ Then } i_o = [20 / (2 + 2)] = 5 \text{ A}, i_1 = i_o / 2 = 2.5 \text{ A}$$

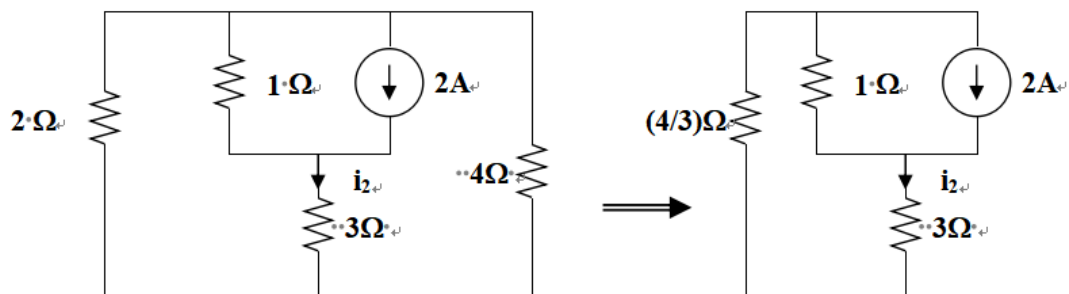
对于 i_3 ,



$$2 \parallel (1+3) = 4/3, \quad v_o' = [(4/3)/((4/3)+4)](-16) = -4$$

$$i_3 = v_o'/4 = -1$$

对于 i_2



$$2 \parallel 4 = 4/3, \quad 3 + 4/3 = 13/3$$

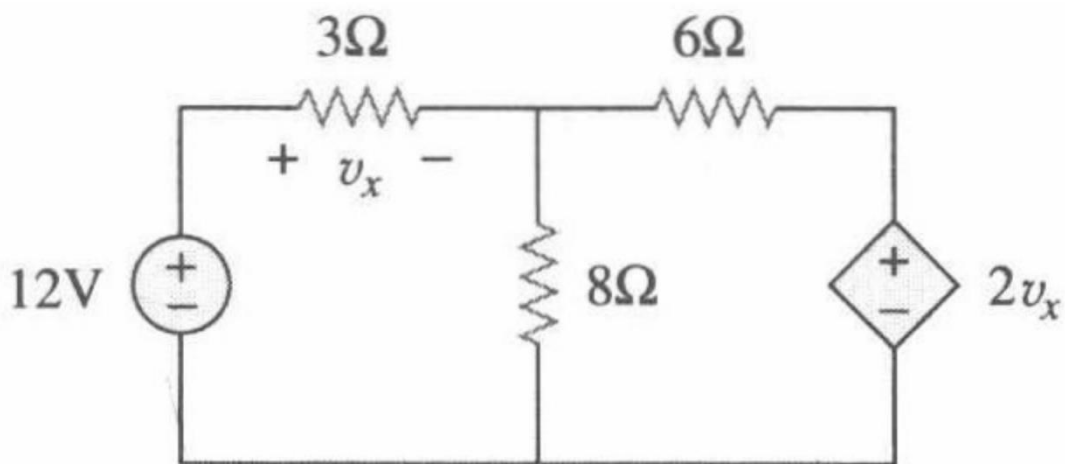
运用分流定理：

$$i_2 = [1/(1+13/2)]2 = 3/8 = 0.375$$

$$i = 2.5 + 0.375 - 1 = 1.875 \text{ A}$$

$$p = i^2 R = (1.875)^2 3 = 10.55 \text{ W}$$

4.31 利用电源变换的方法计算下图所示电路中的 v_x

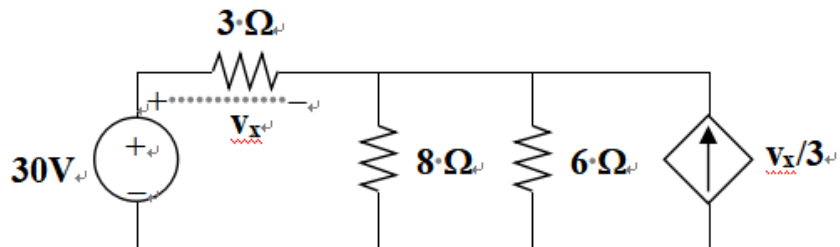


解：

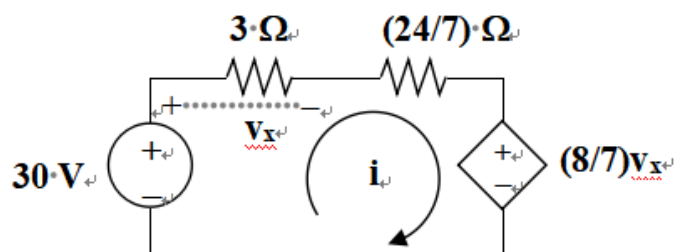
把受控源进行变换可以得到 (a)

$$6 \parallel 8 = (24/7)\Omega$$

把 (a) 中受控源再次变换可以得到 (b)



(a)



(b)

在 (b)

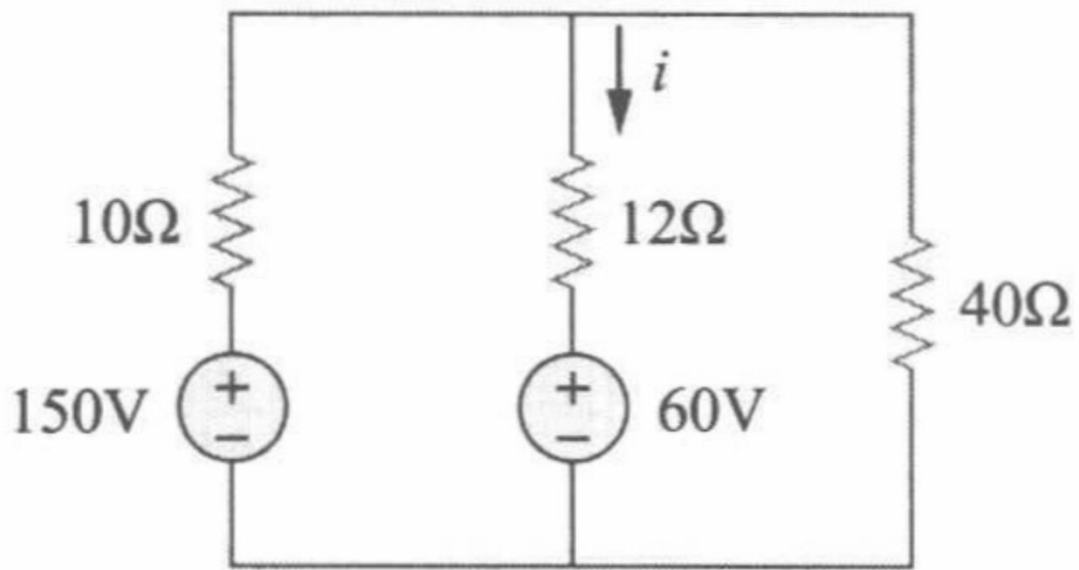
$$v_x = 3i, \quad i = v_x/3.$$

运用 KVL

$$-30 + (3 + 24/7)i + (8/7)v_x = 0$$

$$[(21 + 24)/7]v_x/3 + (8/7)v_x = 30 \text{ leads to } v_x = 30/3.2857 = 9.13 \text{ V.}$$

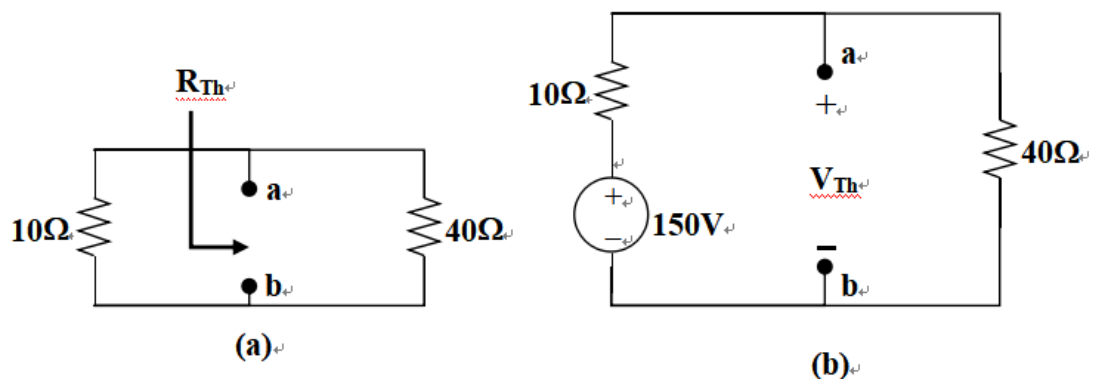
4.36 利用戴维南定理确定下图所示电路中的电流 i (提示：需求出 12Ω 电阻两端的戴维南等效电路)。



解：

移去 60 V 电压源 和 12 Ω 电阻得到(b)，可知如今的戴维南等效电压比计算出来的电压少 60V

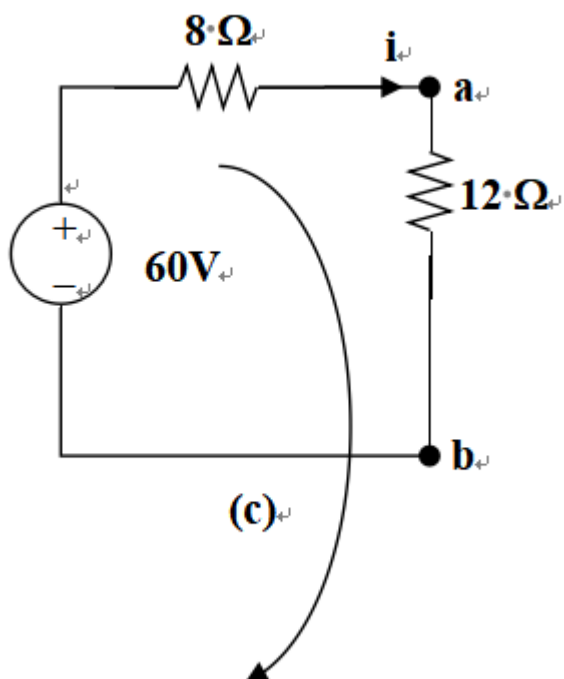
把 150 V 电压源 设为 0得到 (a) .



在(a), $R_{Th} = 10 \parallel 40 = 8 \Omega$

在 (b), $V_{Th} = (40/(10 + 40))150 = 120V$

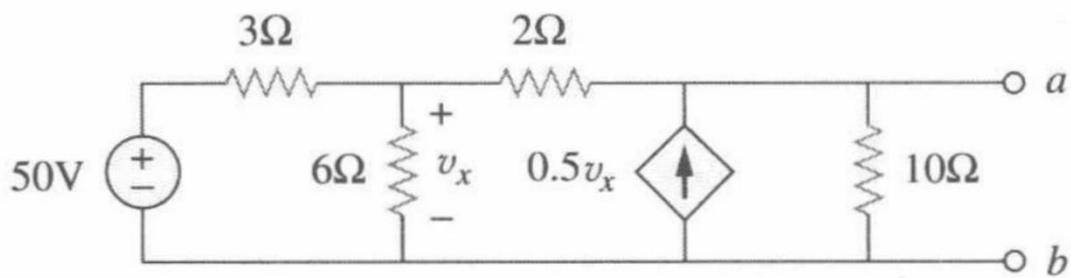
实际的戴维南电压为: $120 - 60 = 60 V$



原始电路的等效电路化为 (c). 运用 KVL,

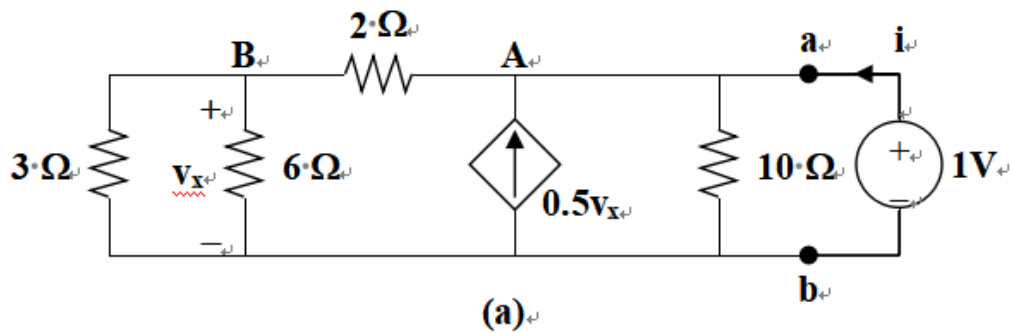
$$\begin{aligned} -60 + (8 + 12)i &= 0 \\ i &= 3 \text{ A.} \end{aligned}$$

4.57 求下图所示电路在端口 a-b 处的戴维南等效电路与诺顿等效电路。



解:

把 50V 电压源移除, 加入一个 1V 电压源可得图 (a)



运用节点分析法，在节点 A

$$i + 0.5v_x = (1/10) + (1 - v_x)/2, \text{ or } i + v_x = 0.6$$

在节点 B,

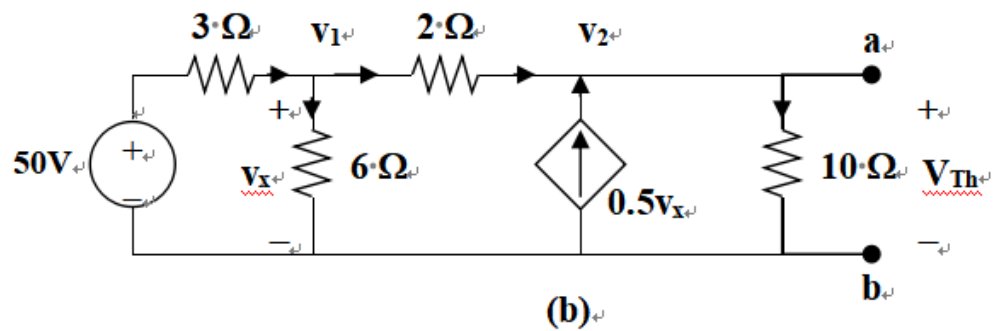
$$(1 - v_o)/2 = (v_x/3) + (v_x/6), \text{ and } v_x = 0.5$$

联立解得

$$i = 0.1$$

$$R_{Th} = 1/i = 10\Omega$$

通过图 (b) 求出 V_{Th}



在节点 1:

$$(50 - v_1)/3 = (v_1/6) + (v_1 - v_2)/2$$

$$100 = 6v_1 - 3v_2$$

在节点 2:

$$0.5v_x + (v_1 - v_2)/2 = v_2/10, \quad v_x = v_1, \text{ and } v_1 = 0.6v_2$$

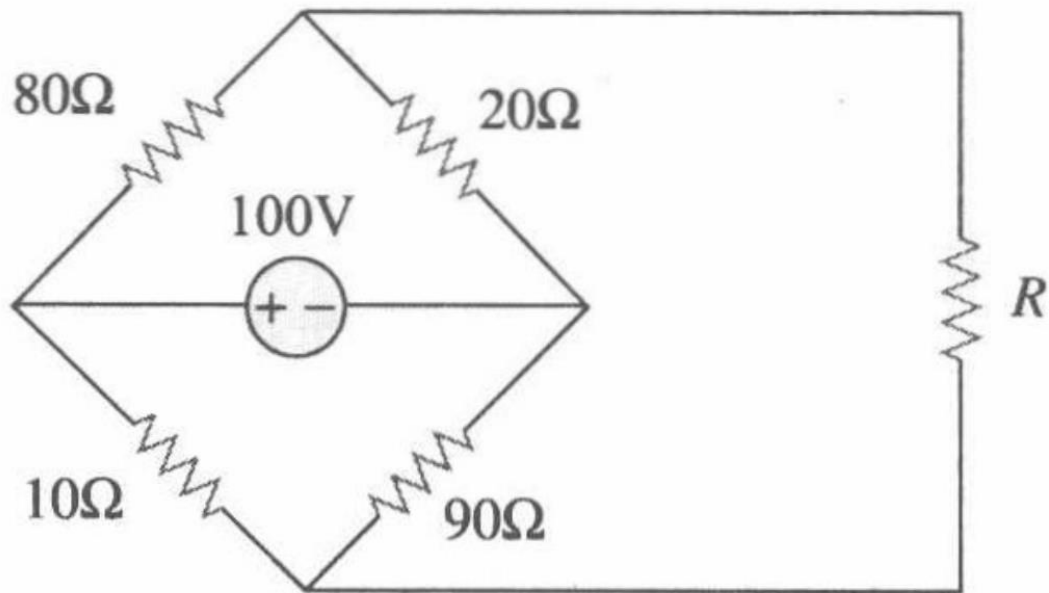
联立解得

$$v_2 = V_{Th} = 166.67 \text{ V}$$

$$I_N = V_{Th}/R_{Th} = 16.667 \text{ A}$$

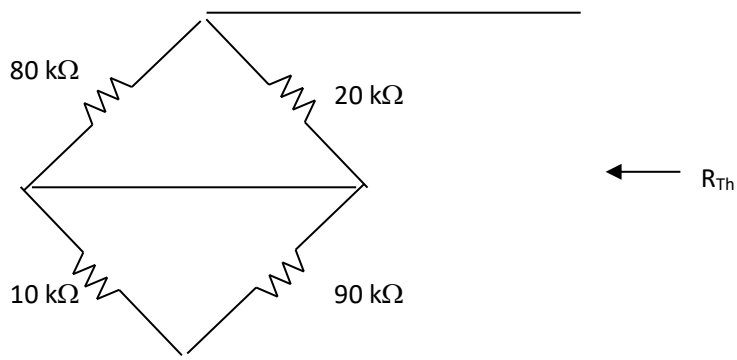
$$R_N = R_{Th} = 10 \Omega$$

4.67 在下图所示电路中，调节可变电阻 R ，直至其从电路中吸收的功率最大。(a) 试计算吸收最大功率时电阻 R 的阻值；(b) 确定 R 吸收的最大功率的值。



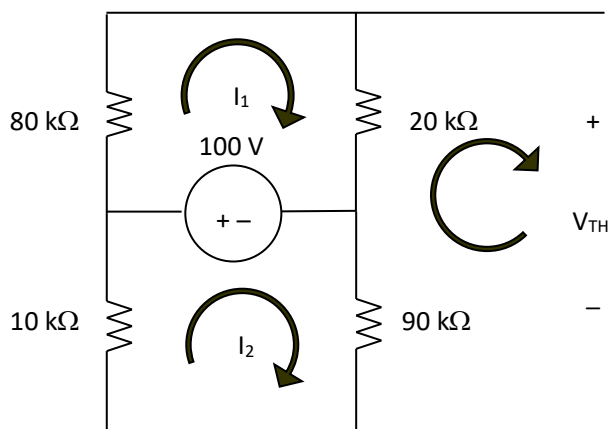
解：

首先得到戴维南等效电路图：



$$R_{Th} = [80k \cdot 20k / (80k + 20k)] + [10k \cdot 90k / (10k + 90k)] = [(1600k / 100) + (900k / 100)] \\ = 16k + 9k = 25k\Omega.$$

然后在原始电路图中运用网孔分析法



网孔 1, $-100 + (80k+20k)i_1 = 0$ or $i_1 = 100/100k = 1 \text{ mA}$.

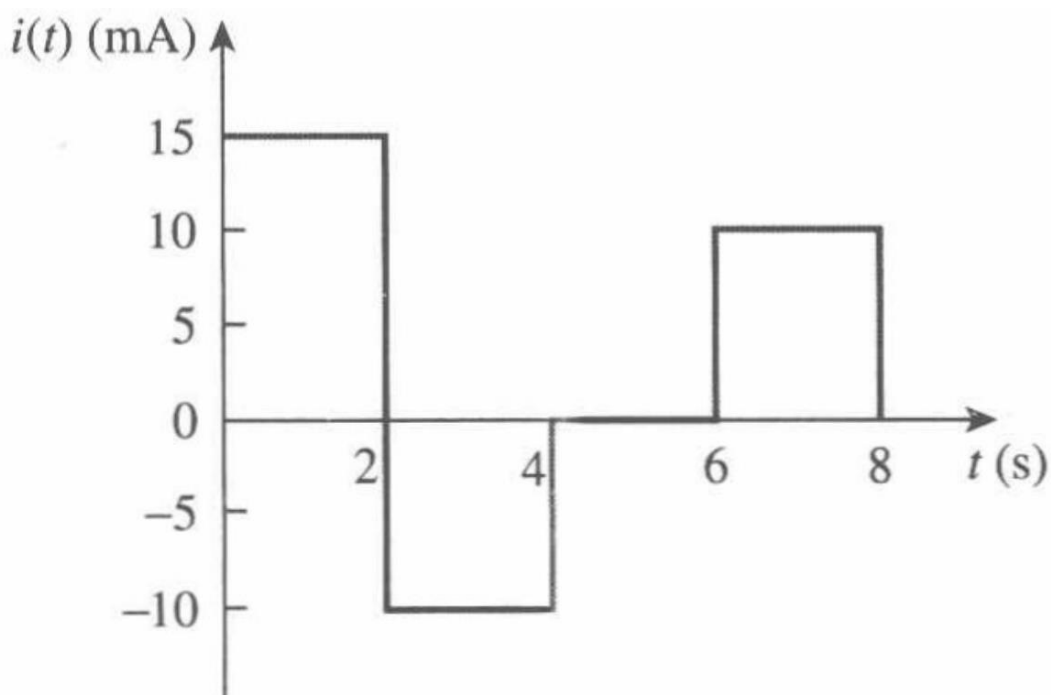
网孔 2, $(10k+90k)i_2 + 100 = 0$ or $i_2 = -100/100k = -1 \text{ mA}$.

最终可得, $V_{\text{Thev}} = 20k(0.001) + 90k(-0.001) = 20-90 = -70 \text{ V}$.

(a) $R = R_{\text{Th}} = 25 \text{ k}\Omega$

(b) $P_{\text{max}} = (V_{\text{Thev}})^2 / (4R_{\text{Thev}}) = (-70)^2 / (4 \times 25k) = 49 \text{ mW}$.

6-11 如果通过一个 4mF 电容的电流波形如下图所示, 假设 $v(0) = 10\text{V}$, 画出电压 $v(t)$ 的波形。



解:

$$v = \frac{1}{C} \int_0^t i dt + v(0) = 10 + \frac{1}{4 \times 10^{-3}} \int_0^t i(t) dt$$

$$0 < t < 2, \quad i(t) = 15 \text{ mA}, \quad v(t) = 10 + \frac{10^3}{4 \times 10^{-3}} \int_0^t 15 dt = 10 + 3.75t$$

$$v(2) = 10 + 7.5 = 17.5$$

$$2 < t < 4, \quad i(t) = -10 \text{ mA}$$

$$v(t) = \frac{1}{4 \times 10^{-3}} \int_2^t i(t) dt + v(2) = -\frac{10 \times 10^{-3}}{4 \times 10^{-3}} \int_2^t dt + 17.5 = 22.5 - 2.5t$$

$$v(4)=22.5-2.5 \times 4=12.5$$

$$4 < t < 6, \quad i(t) = 0, \quad v(t) = \frac{1}{4 \times 10^{-3}} \int_2^t 0 dt + v(4) = 12.5$$

$$6 < t < 8, \quad i(t) = 10 \text{ mA}$$

$$v(t) = \frac{10 \times 10^{-3}}{4 \times 10^{-3}} \int_4^t dt + v(6) = 2.5(t - 6) + 12.5 = 2.5t - 2.5$$

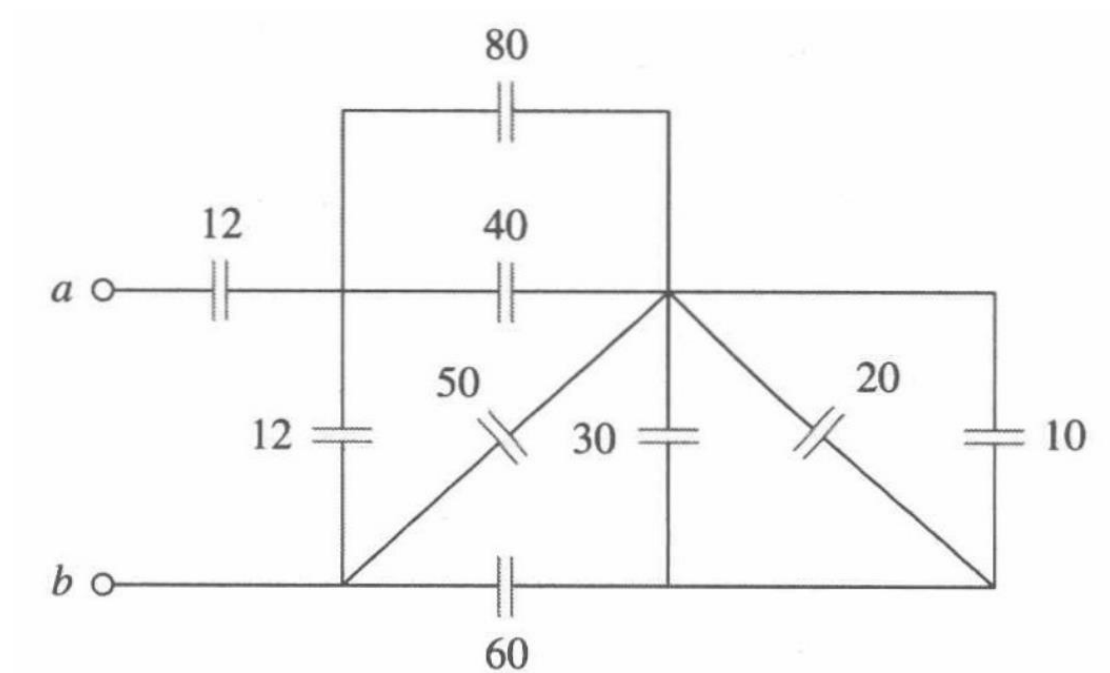
所以,

$$v(t) = \begin{cases} 10 + 3.75t \text{ V}, & 0 < t < 2 \text{ s} \\ 22.5 - 2.5t \text{ V}, & 2 < t < 4 \text{ s} \\ 12.5 \text{ V}, & 4 < t < 6 \text{ s} \\ 2.5t - 2.5 \text{ V}, & 6 < t < 8 \text{ s} \end{cases}$$

画出图像:



6-19 计算下图所示电路终端 a 和 b 间的等效电容, 所有电容单位均为 μF 。

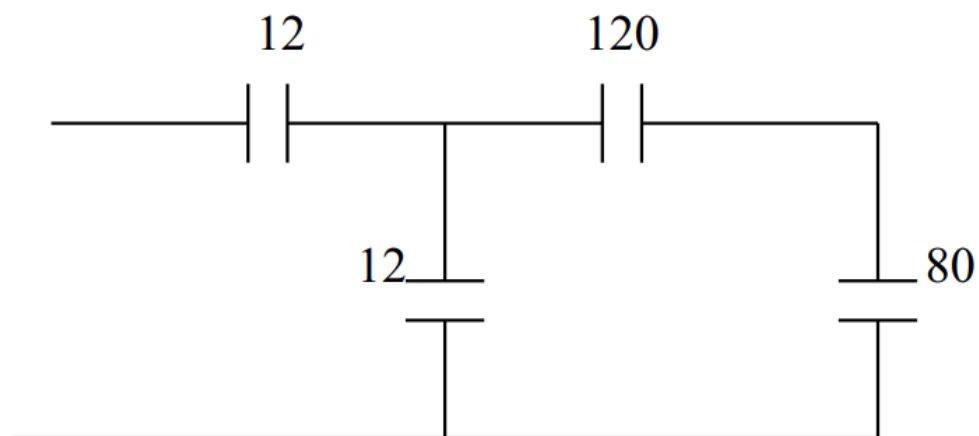


解:

把 10-, 20-, 30- μF 的电容进行平行等效可以得到 60 μF . 60- μF 电容与 60- μF 电容串联等效得到 30 μF .

$30 + 50 = 80 \mu\text{F}$, $80 + 40 = 120 \mu\text{F}$

可以得到下面的电路:

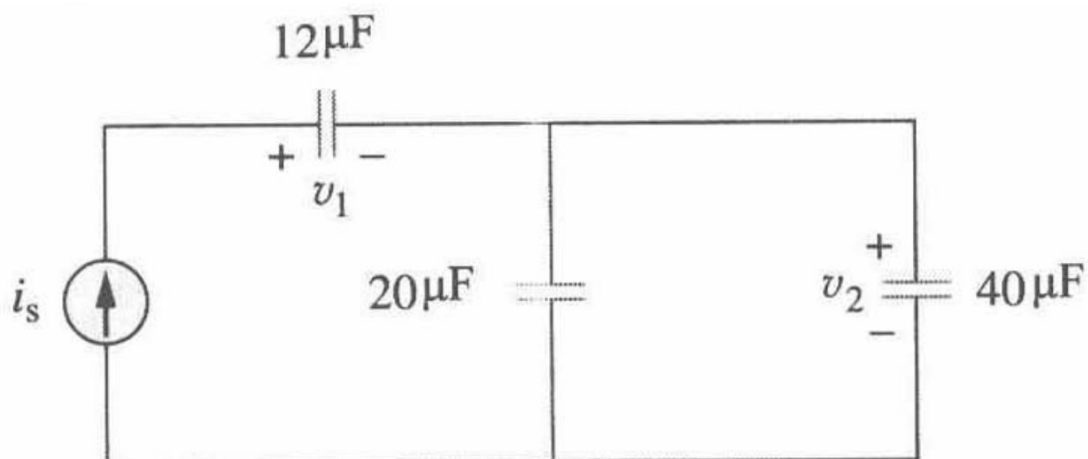


120- μF 电容与 80 μF 电容进行串联等效 $(80 \times 120) / 200 = 48$

$48 + 12 = 60$

60- μF 电容与 12 μF 电容进行串联等效 $(60 \times 12) / 72 = 10 \mu\text{F}$

6-32 在下图所示电路中, $i_s = 4.5e^{-2t} \text{mA}$ 且当 $t=0$ 时, $v_1(0) = 0\text{V}$, $v_2(0) = 0\text{V}$ 。计算: (a) $v_1(t)$, $v_2(t)$; (b) 在 $t > 0\text{s}$ 时每个电容所存储的能量。



解:

把 $20\mu\text{F}$ 电容与 $40\mu\text{F}$ 电容并联合并可以得到 $60\mu\text{F}$

$$v_1 = \frac{1}{60\mu} \int_0^t 4.5e^{-2\tau} m d\tau = [37.5 - 37.5e^{-2t}] \text{ V}$$

$$v_2 = [187.5 - 187.5e^{-2t}] \text{ V}$$

$$(v_1)^2 = 35156.25[1 - 2e^{-2t} + e^{-4t}]$$

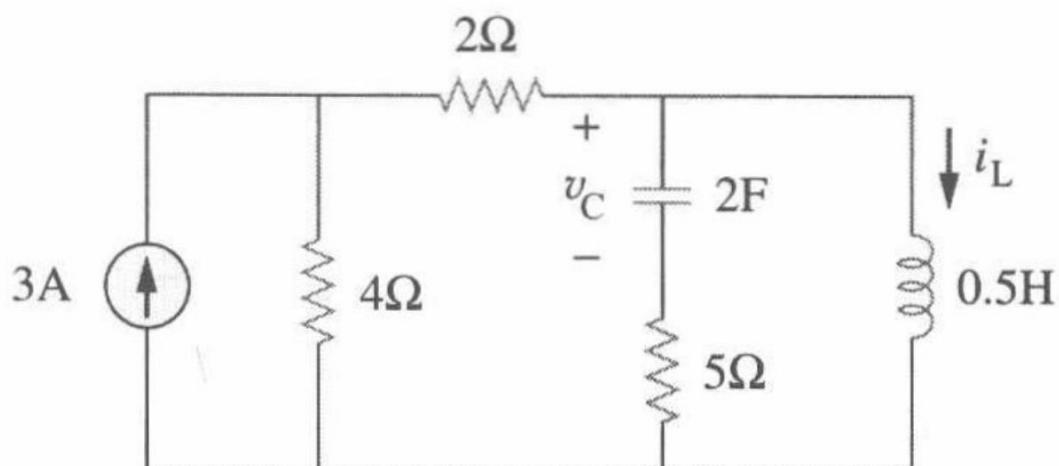
$$(v_2)^2 = 1406.25[1 - 2e^{-2t} + e^{-4t}]$$

$$W_1 = 0.5 \times 12 \times 10^{-6} (v_1)^2 = \mathbf{0.211[1 - 2e^{-2t} + e^{-4t}]} \text{ J}$$

$$W_2 = 0.5 \times 20 \times 10^{-6} (v_2)^2 = \mathbf{0.01406[1 - 2e^{-2t} + e^{-4t}]} \text{ J}$$

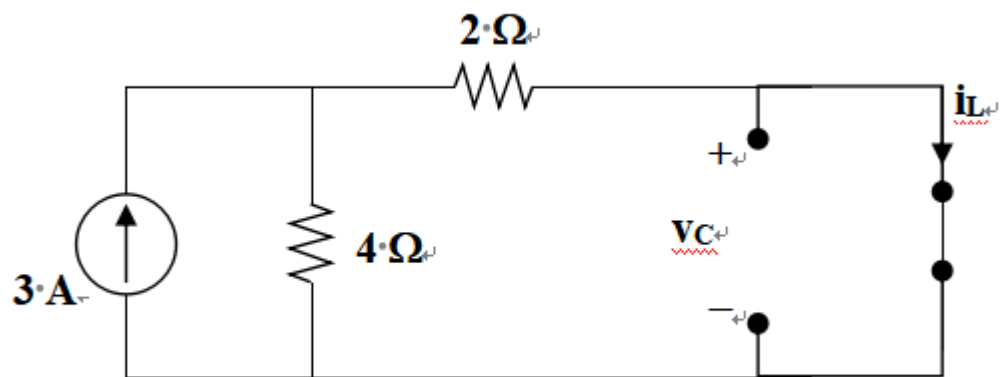
$$W_3 = 0.5 \times 40 \times 10^{-6} (v_2)^2 = \mathbf{0.02813[1 - 2e^{-2t} + e^{-4t}]} \text{ J}$$

6-46 在下图所示直流电路中，计算电容上的电压 v_c ，电感上的电流 i_L ，以及它们分别所储存的能量。



解:

在直流情况下，电路如下图所示



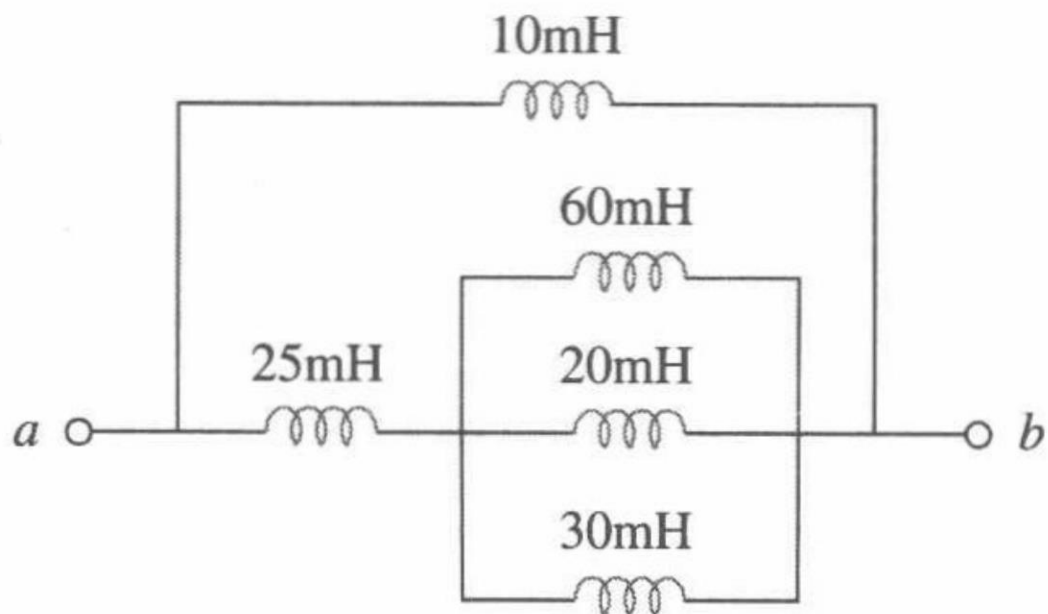
运用分流定理

$$i_L = \frac{4}{4+2}(3) = 2\text{A}, \quad v_c = 0\text{V}$$

$$w_L = \frac{1}{2} L i_L^2 = \frac{1}{2} \left(\frac{1}{2} \right) (2)^2 = 1\text{J}$$

$$w_c = \frac{1}{2} C v_c^2 = \frac{1}{2} (2)(0) = 0\text{J}$$

6-51 计算下图所示电路 a-b 终端间的等效电感 L_{eq} 。

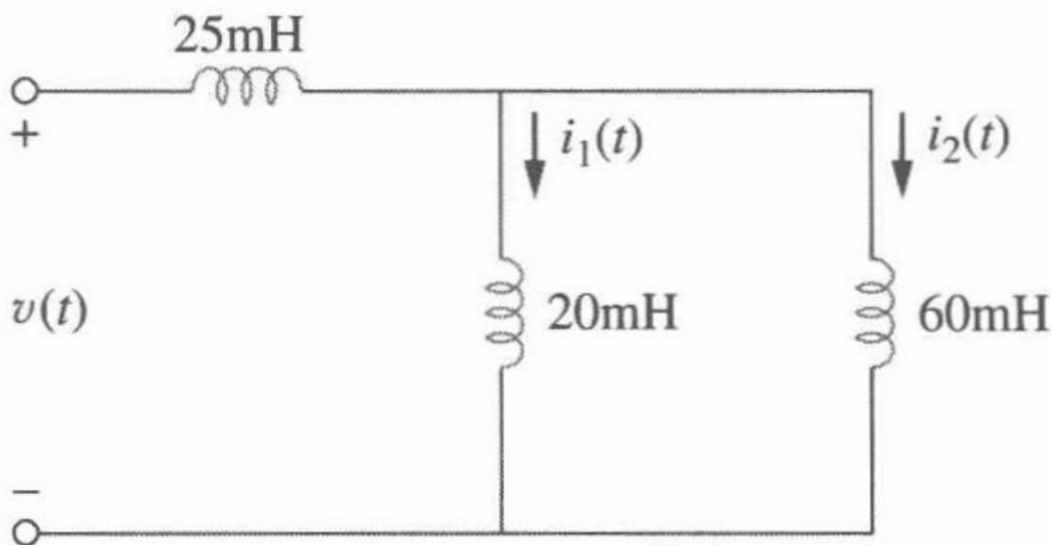


解：

$$\frac{1}{L} = \frac{1}{60} + \frac{1}{20} + \frac{1}{30} = \frac{1}{10} \quad L = 10 \text{ mH}$$

$$L_{eq} = 10 \parallel (25 + 10) = \frac{10 \times 35}{45} = 7.778 \text{ mH}$$

6-62 分析下图所示电路，在 $t > 0$ 、 $v(t) = 12e^{-3t} \text{ mV}$ 、 $i_1(0) = -30 \text{ mA}$ 时，计算：(a) $i_2(0)$ ；
(b) $i_1(t)$ 和 $i_2(t)$ 。



解：

$$(a) \quad L_{eq} = 25 + 20 \parallel 60 = 25 + \frac{20 \times 60}{80} = 40 \text{ mH}$$

$$v = L_{eq} \frac{di}{dt} \longrightarrow i = \frac{1}{L_{eq}} \int v(t) dt + i(0) = \frac{10^{-3}}{40 \times 10^{-3}} \int_0^t 12e^{-3t} dt + i(0) = -0.1(e^{-3t} - 1) + i(0)$$

运用分流定理

$$i_1 = \frac{60}{80} i = \frac{3}{4} i, \quad i_2 = \frac{1}{4} i$$

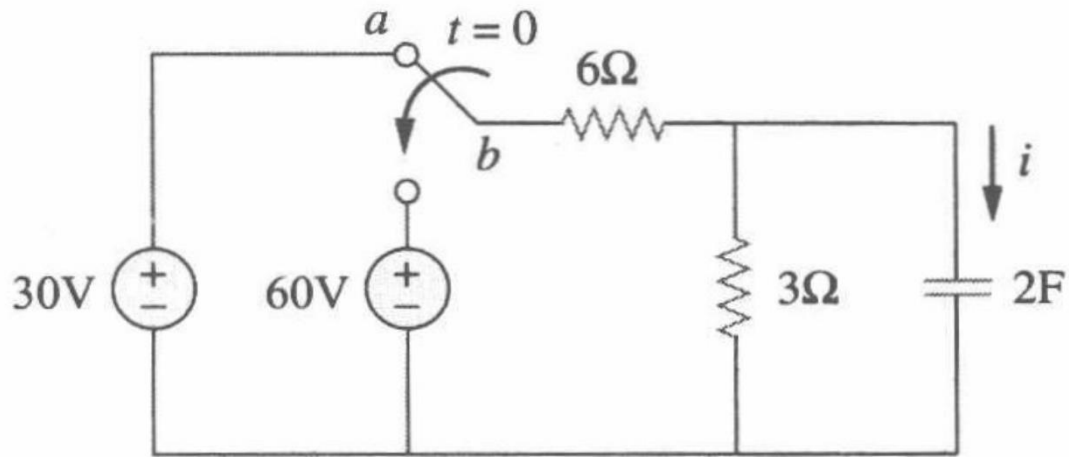
$$i_1(0) = \frac{3}{4} i(0) \longrightarrow 0.75 i(0) = -0.03 \longrightarrow i(0) = -0.04$$

$$i_2 = \frac{1}{4} (-0.1e^{-3t} + 0.06) \text{ A} = (-25e^{-3t} + 15) \text{ mA}$$

$$i_2(0) = -25 + 15 = -10 \text{ mA.}$$

$$(b) \quad i_1(t) = 0.75(-0.1e^{-3t} + 0.06) = (-75e^{-3t} + 45) \text{ mA} \text{ and } i_2(t) = (-25e^{-3t} + 15) \text{ mA.}$$

7.44 下图所示电路中，开关处于 a 位置且电路已达到稳态，当 $t=0$ 时，开关切换到 b 位置，求 $t>0$ 时的 $i(t)$ 。



解：

$$R_{eq} = 6 \parallel 3 = 2 \Omega$$

$$\tau = RC = 4$$

$$v(t) = v(\infty) + [v(0) - v(\infty)] e^{-t/\tau}$$

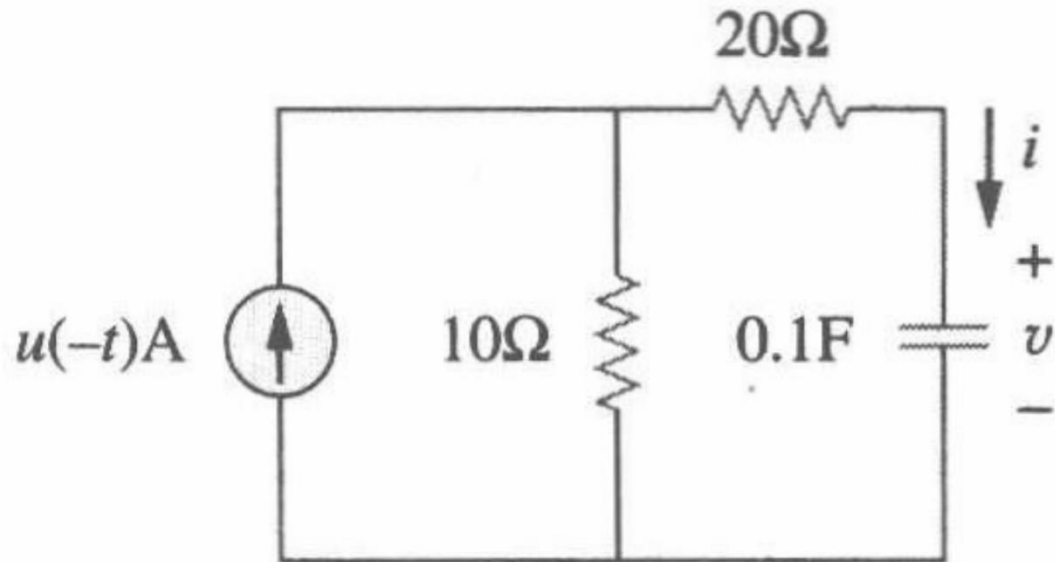
运用分压定理,

$$v(0) = \frac{3}{3+6} (30) = 10 \text{ V}, \quad v(\infty) = \frac{3}{3+6} (60) = 20 \text{ V}$$

$$v(t) = 20 + (10 - 20) e^{-t/4} = 20 - 10 e^{-t/4}$$

$$i(t) = C \frac{dv}{dt} = (2)(-10)(-1/4) e^{-t/4} = (5e^{-0.25t})u(t) \text{ A.}$$

7.48 下图所示电路中，求 $v(t)$ 和 $i(t)$



解：

当 $t < 0$, $u(-t) = 1$,

当 $t > 0$, $u(-t) = 0$, $v(\infty) = 0$

$$R_{th} = 20 + 10 = 30, \quad \tau = R_{th}C = (30)(0.1) = 3$$

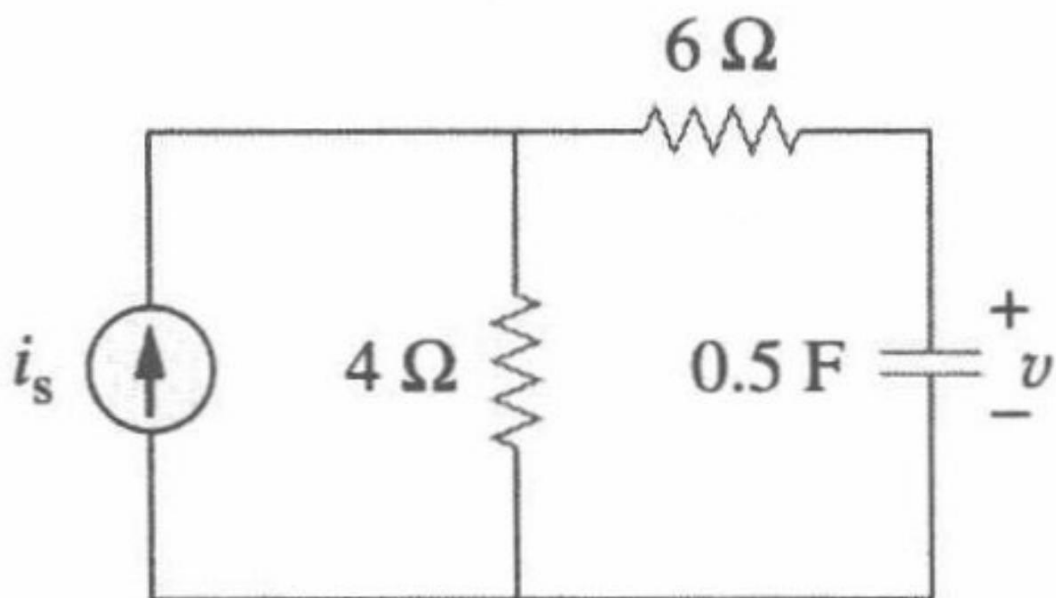
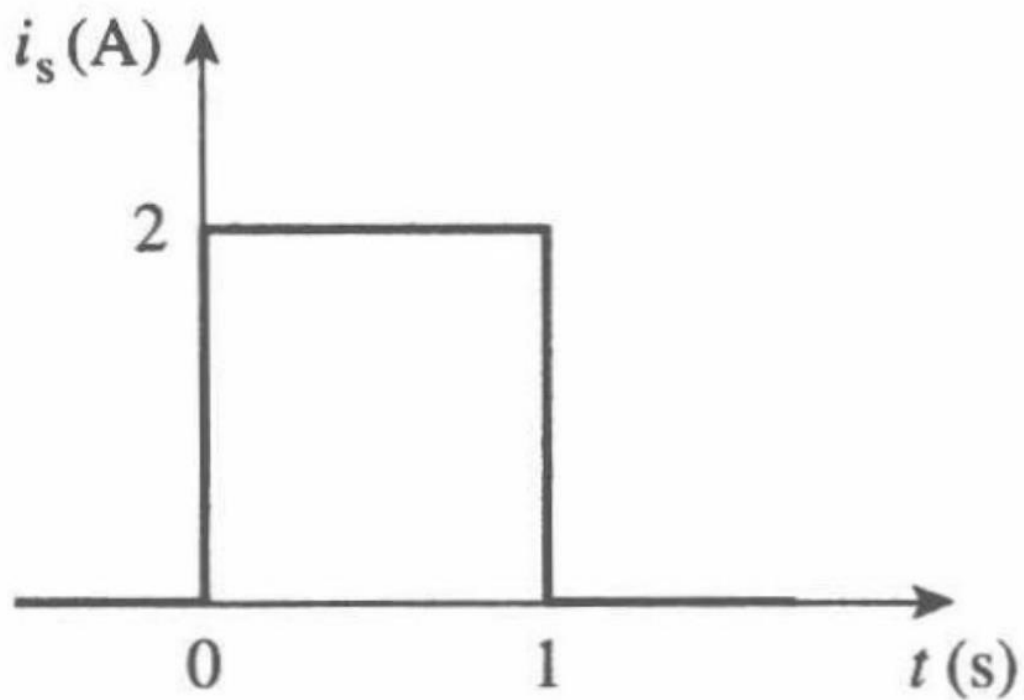
$$v(t) = v(\infty) + [v(0) - v(\infty)] e^{-t/\tau}$$

$$v(t) = 10e^{-t/3} \text{ V}$$

$$i(t) = C \frac{dv}{dt} = (0.1) \left(\frac{-1}{3} \right) 10e^{-t/3}$$

$$i(t) = \frac{-1}{3} e^{-t/3} \text{ A}$$

7.49 下图所示波形对对应下图所示电路，假设 $v(0) = 0$ ，求 $v(t)$



解：

当 $0 < t < 1$, $v(0) = 0$, $v(\infty) = (2)(4) = 8$

$$R_{eq} = 4 + 6 = 10, \quad \tau = R_{eq}C = (10)(0.5) = 5$$

$$v(t) = v(\infty) + [v(0) - v(\infty)] e^{-t/\tau}$$

$$v(t) = 8(1 - e^{-t/5}) \text{ V}$$

当 $t > 1$, $v(1) = 8(1 - e^{-0.2}) = 1.45$, $v(\infty) = 0$

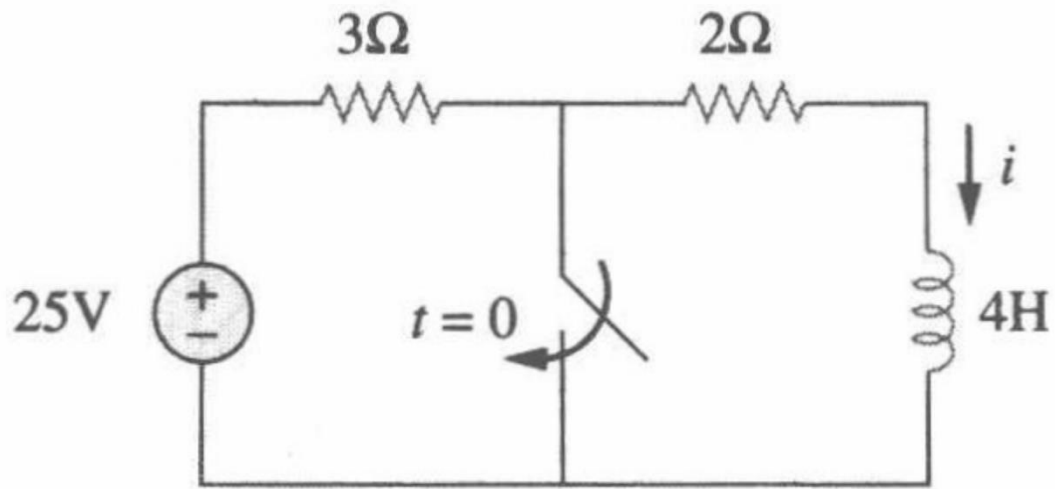
$$v(t) = v(\infty) + [v(1) - v(\infty)] e^{-(t-1)/\tau}$$

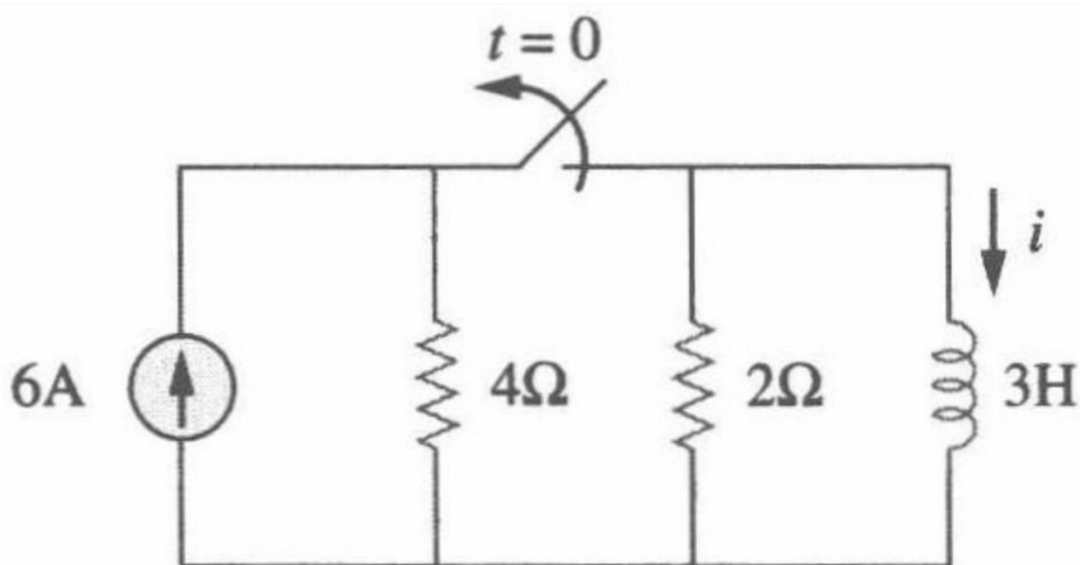
$$v(t) = 1.45 e^{-(t-1)/5} \text{ V}$$

因此,

$$v(t) = \begin{cases} 8(1 - e^{-t/5}) \text{ V}, & 0 < t < 1 \\ 1.45 e^{-(t-1)/5} \text{ V}, & t > 1 \end{cases}$$

7.53 下图中, 求每个电路在 $t < 0$ 和 $t > 0$ 时的电感电流 $i(t)$ 。





(a) $t = 0$ 之前, $i = \frac{25}{3+2} = 5 \text{ A}$

$t = 0$ 之后, $i(t) = i(0)e^{-t/\tau}$

$\tau = \frac{L}{R} = \frac{4}{2} = 2, i(0) = 5$

$i(t) = 5e^{-t/2} u(t) \text{ A}$

(b) $t = 0$ 之前,

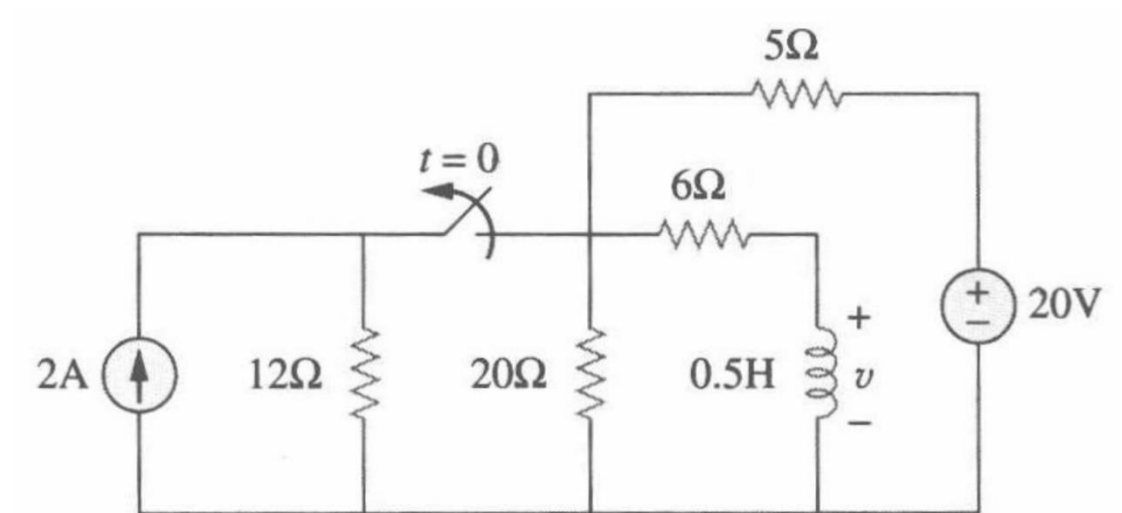
$i(t) = 6 \text{ A}$

$t = 0$ 之后,

$i(t) = i(0)e^{-t/\tau}, \quad \tau = \frac{L}{R} = \frac{3}{2}$

$i(t) = 6e^{-2t/3} u(t) \text{ A}$

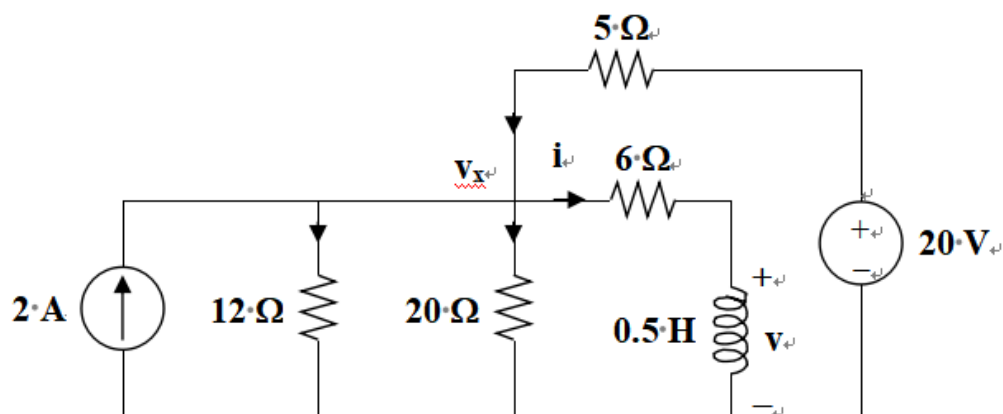
7.56 在下图所示网络中, 求当 $t > 0$ 时的 $v(t)$ 。



解:

$$R_{eq} = 6 + 20 \parallel 5 = 10 \Omega, \quad \tau = \frac{L}{R} = 0.05$$

$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$



$$2 + \frac{20 - v_x}{5} = \frac{v_x}{12} + \frac{v_x}{20} + \frac{v_x}{6} \longrightarrow v_x = 12$$

$$i(0) = \frac{v_x}{6} = 2 \text{ A}$$

因为 $20 \parallel 5 = 4$,

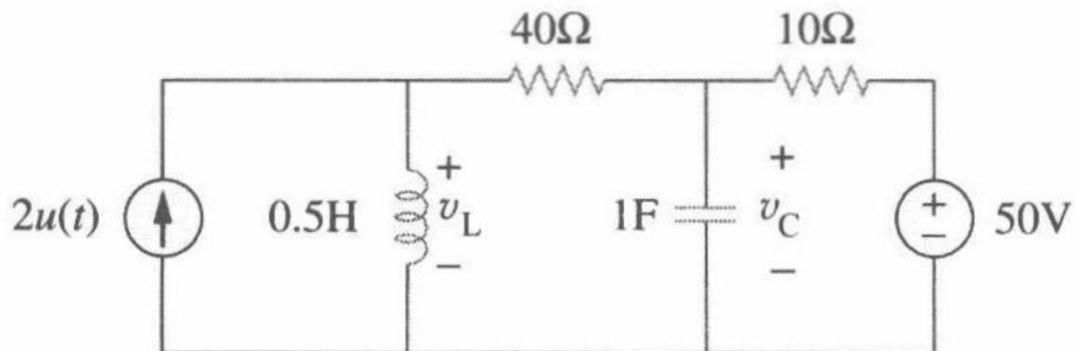
$$i(\infty) = \frac{4}{4 + 6} (4) = 1.6$$

$$i(t) = 1.6 + (2 - 1.6)e^{-t/0.05} = 1.6 + 0.4e^{-20t}$$

$$v(t) = L \frac{di}{dt} = \frac{1}{2} (0.4) (-20) e^{-20t}$$

$$v(t) = -4e^{-20t} \text{ V}$$

8.31 对于下图所示电路，计算 $v_L(0^+)$ 以及 $v_C(0^+)$ 。



解：

在 $t = 0^-$ ，可以得到电路图 (a)。在 $t = 0^+$ ，可以得到电路图 (b)。运用 KVL,

$$v(0^+) = v(0^-) = 40, \quad i(0^+) = i(0^-) = 1$$

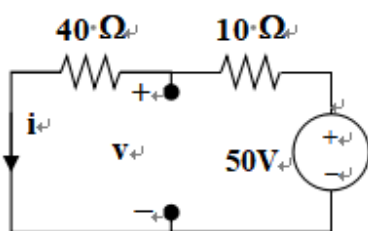
运用 KCL, $2 = i(0^+) + i_1 = 1 + i_1$

$$i_1 = 1.$$

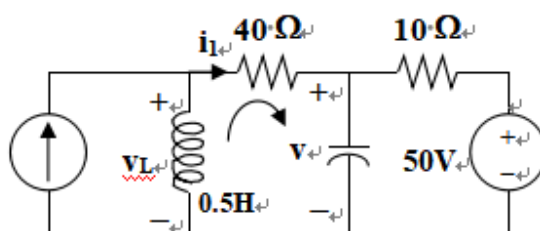
运用 KVL, $-v_L + 40i_1 + v(0^+) = 0$

$$v_L(0^+) = 40 \times 1 + 40 = 80$$

$$v_L(0^+) = 80 \text{ V}, \quad v_C(0^+) = 40 \text{ V}$$

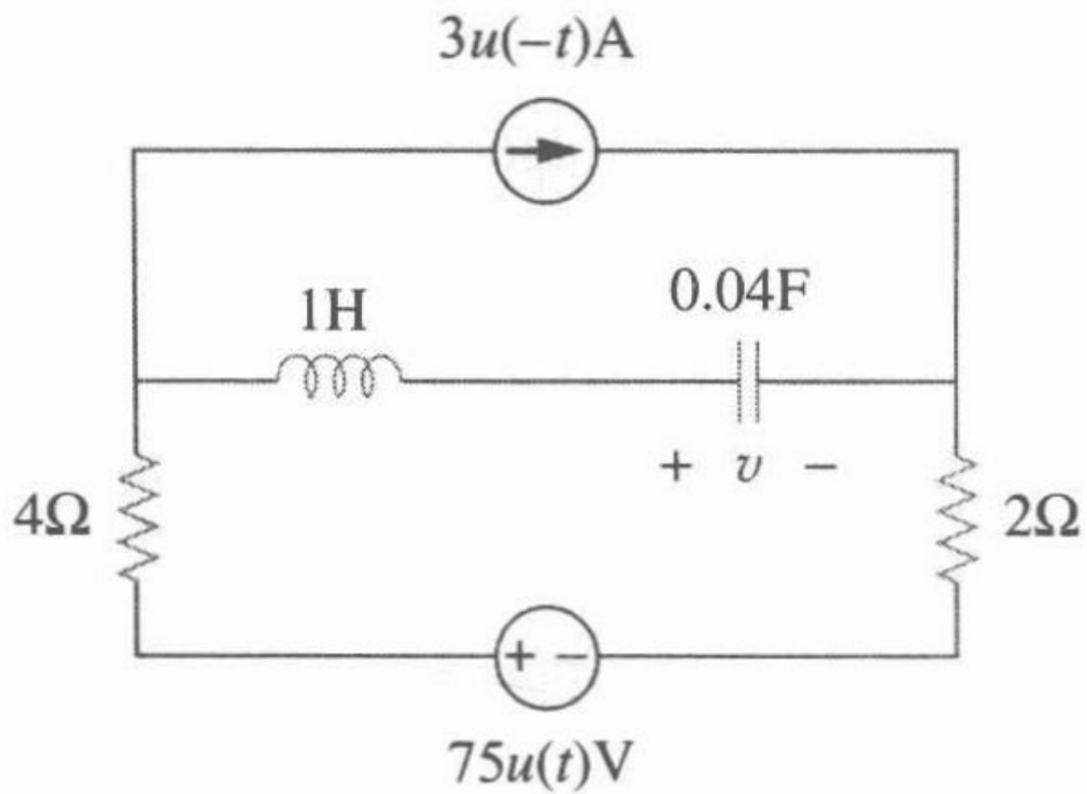


(a)



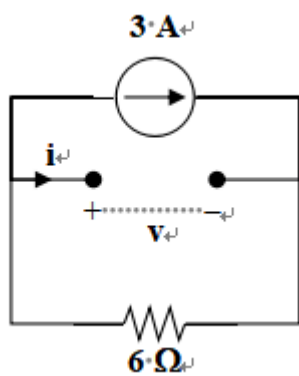
(b)

8.32 对于下图所示电路，计算 $t > 0$ 时的 $v(t)$



解:

在 $t = 0^-$, 等效电路图为:



$$i(0^-) = 0, \quad v(0^-) = -3 \times 6 = -18 \text{ V}$$

在 $t > 0$,

$$\alpha = R/(2L) = 6/2 = 3, \quad \omega_0 = 1/\sqrt{LC} = 1/\sqrt{0.04}$$

$$s_{1,2} = -3 \pm \sqrt{9 - 25} = -3 \pm j4$$

因此, $v(t) = V_s + [(A\cos 4t + B\sin 4t)e^{-3t}]$

$$V_s = 75 \text{ V}$$

$$v(t) = 75 + [(A\cos 4t + B\sin 4t)e^{-3t}]$$

$$v(0) = -18 = 75 + A \quad \text{得到} \quad A = -93.$$

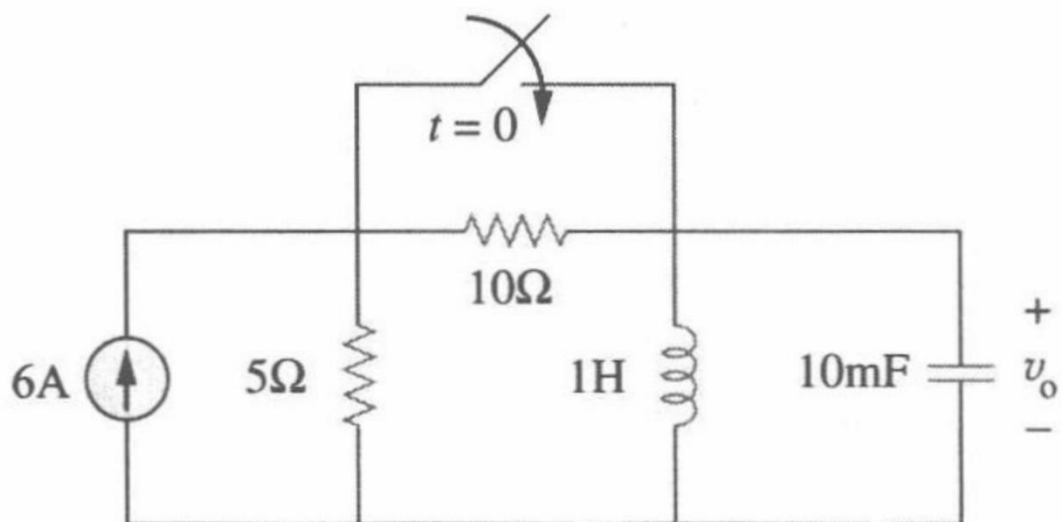
$$i(0) = 0 = Cdv(0)/dt$$

$$dv/dt = [-3(A\cos 4t + B\sin 4t)e^{-3t}] + [4(-A\sin 4t + B\cos 4t)e^{-3t}]$$

$$0 = dv(0)/dt = -3A + 4B \quad \text{or} \quad B = (3/4)A = -69.75$$

$$v(t) = \{75 + [(-93\cos 4t - 69.75\sin 4t)e^{-3t}]\} \text{ V}, \quad t > 0.$$

8.47 计算下图所示电路的输出电压 $v_o(t)$



解:

$$\text{在 } t = 0^-, \quad i_L(0) = 6 \times 5 / (10 + 5) = 2 \text{ A}, \quad v_o(0) = 0.$$

在 $t > 0$,

$$\alpha = 1/(2RC) = (1)/(2 \times 5 \times 0.01) = 10$$

$$\omega_o = 1/\sqrt{LC} = 1/\sqrt{1 \times 0.01} = 10$$

因为 $\alpha = \omega_o$,

$$s_{1,2} = -10$$

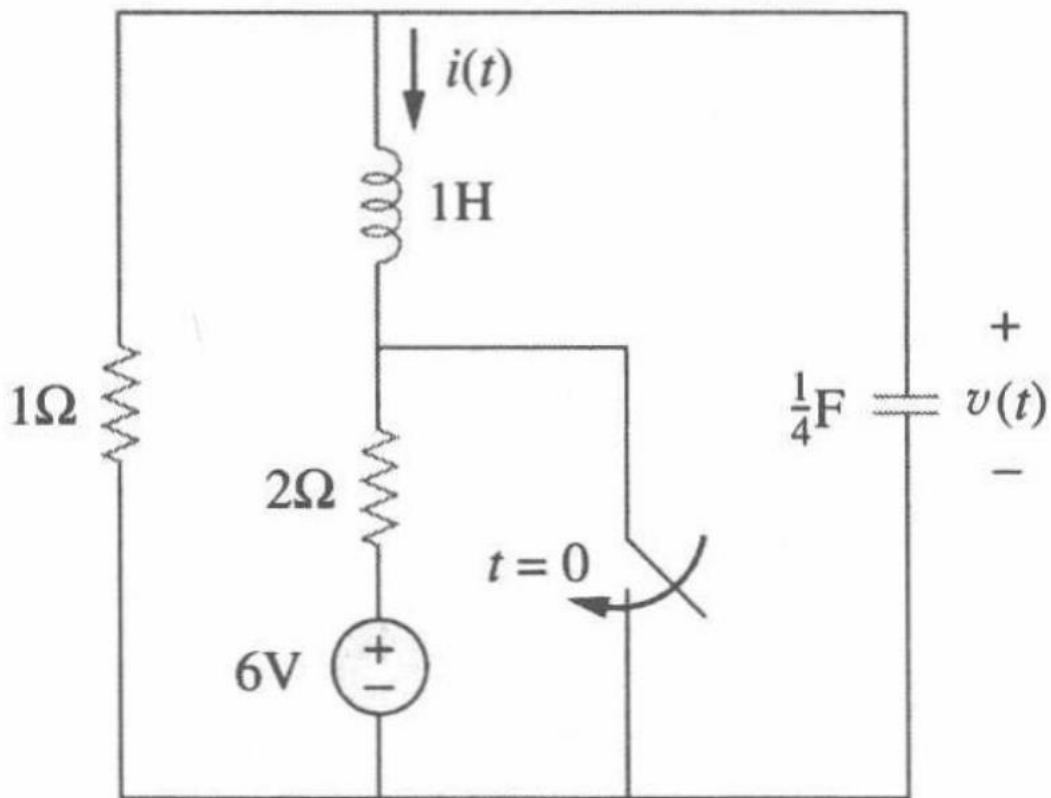
因此 $i(t) = I_{ss} + [(A + Bt)e^{-10t}]$, $I_{ss} = 6$, $i(0) = 2 = 6 + A$, $A = -4$

$$v_o = L di/dt = [Be^{-10t}] + [-10(A + Bt)e^{-10t}]$$

$$v_o(0) = 0 = B - 10A, \quad B = -40$$

因此, $v_o(t) = (400te^{-10t}) \text{ V}$.

8.48 对于下图所示电路，计算 $t > 0$ 时的 $v(t)$ 和 $i(t)$ 。



解:

在 $t = 0^-$, $i(0) = -6/(1+2) = -2$, $v(0) = 2 \times 1 = 2$.

在 $t > 0$,

$$\alpha = 1/(2RC) = (1)/(2 \times 1 \times 0.25) = 2$$

$$\omega_0 = 1/\sqrt{LC} = 1/\sqrt{1 \times 0.25} = 2$$

因为 $\alpha = \omega_0$,

$$s_{1,2} = -2$$

因此, $i(t) = [(A + Bt)e^{-2t}]$, $i(0) = -2 = A$

$$v = L di/dt = [Be^{-2t}] + [-2(-2 + Bt)e^{-2t}]$$

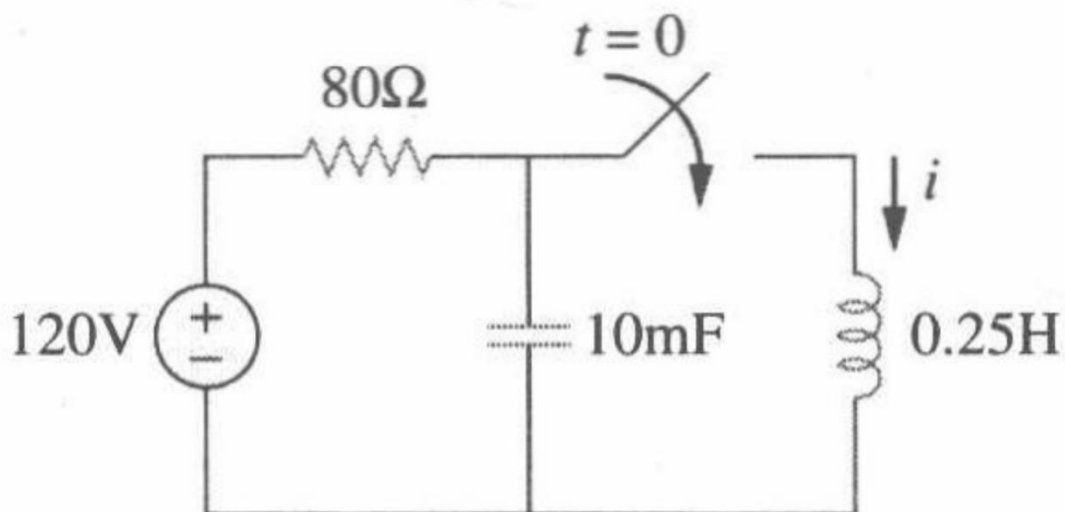
$$v_0(0) = 2 = B + 4 \text{ or } B = -2$$

因此,

$$i(t) = [(-2 - 2t)e^{-2t}] \text{ A}$$

$$v(t) = [(2 + 4t)e^{-2t}] \text{ V}$$

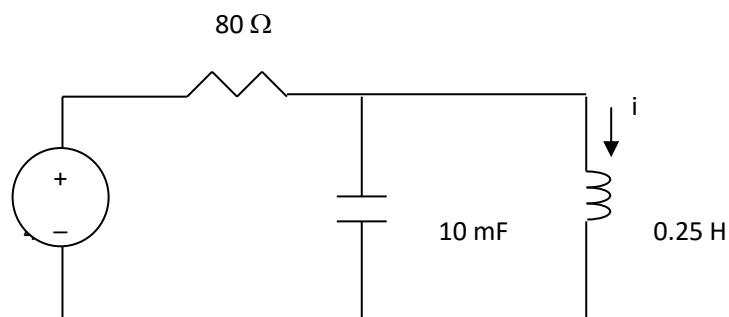
8.53 下图所示电路中, 已断开一天的开关在 $t=0$ 时闭合。计算 $t>0$ 时 $i(t)$ 的差分方程。



解:

当 $t < 0$, $i(0) = 0$, $v_C(0) = 40$.

当 $t > 0$,



$$[(40 - v_C)/80] = 0.01[dv_C/dt] + i, \quad 40 = v_C + 0.8[dv_C/dt] + 80i.$$

$$(d^2i/dt^2) + 1.25(di/dt) + 400i = 200.$$

9.5 已知 $v_1 = 45\sin(\omega t + 30^\circ)$ V 和 $v_2 = 50\cos(\omega t - 30^\circ)$ V, 计算这两个正弦信号之间的相位角, 并指出哪一个是滞后的。

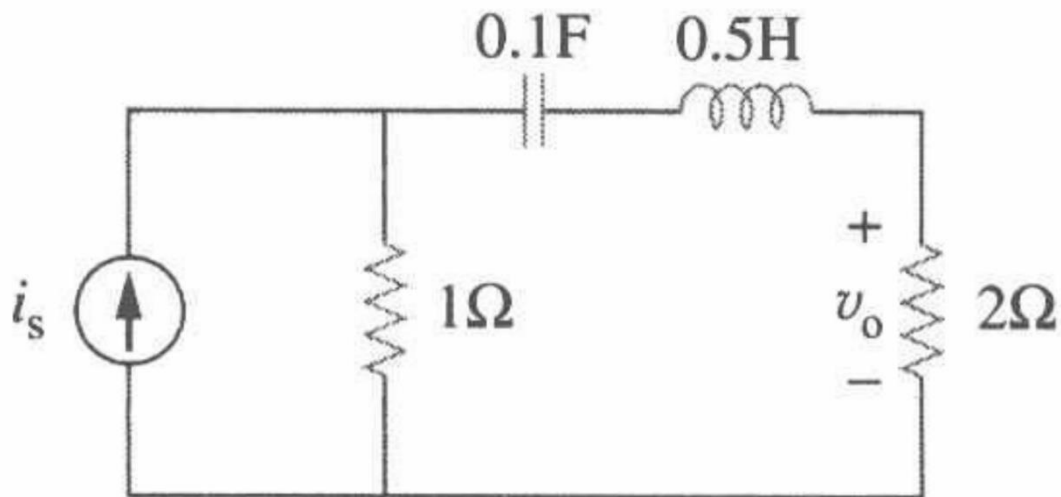
解:

$$v_1 = 45\sin(\omega t + 30^\circ) \text{ V} = 45\cos(\omega t + 30^\circ - 90^\circ) = 45\cos(\omega t - 60^\circ) \text{ V}$$

$$v_2 = 50\cos(\omega t - 30^\circ) \text{ V}$$

相位角为 30° , v_1 滞后于 v_2 .

9.51 如果下图所示电路中 2Ω 电阻两端的电压 v_o 为 $90\cos 2t$ V, 试求 i_s .



解:

$$0.1 \text{ F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(2)(0.1)} = -j5$$

$$0.5 \text{ H} \longrightarrow j\omega L = j(2)(0.5) = j$$

2-Ω 电阻的电路为

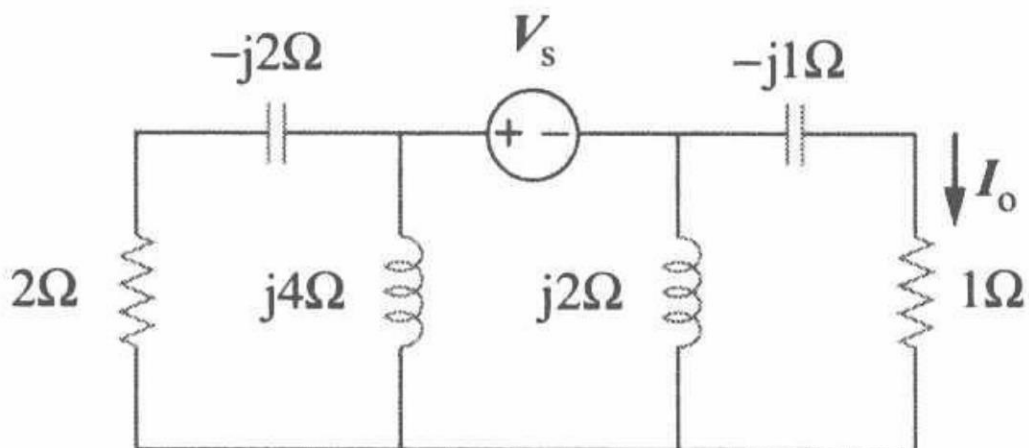
$$\mathbf{I} = \frac{1}{1 - j5 + j + 2} \mathbf{I}_s = \frac{\mathbf{I}_s}{3 - j4}, \quad \mathbf{I} = \frac{90}{2} \angle 0^\circ = 45$$

$$\mathbf{I}_s = (45)(3 - j4) = 225 \angle -53.13^\circ$$

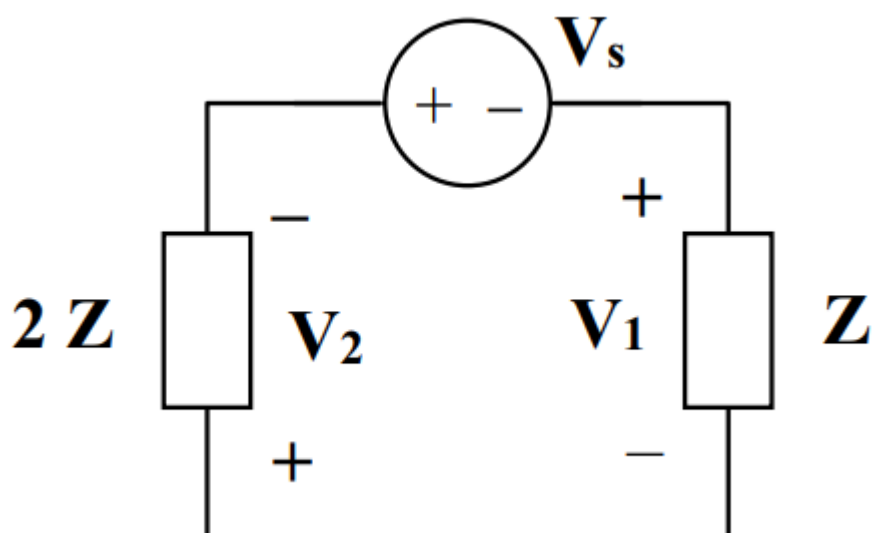
因此,

$$i_s(t) = 225 \cos(2t - 53.13^\circ) \text{ A}$$

9.54 在下图所示电路中, $\mathbf{I}_o = 30 \angle 0^\circ \text{ A}$, 试求 $\mathbf{V}_s$ 。



解：
等效电路如图：



$$\mathbf{V}_1 = \mathbf{I}_o(1 - j) = 30(1 - j)$$

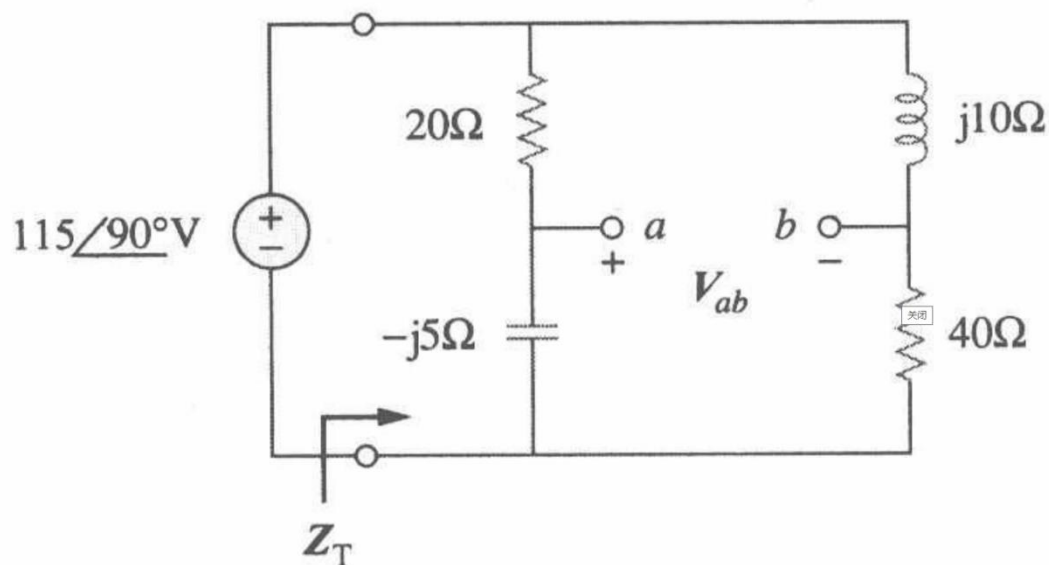
$$\mathbf{V}_2 = 2\mathbf{V}_1 = 60(1 - j)$$

$$\mathbf{V}_2 + \mathbf{V}_s + \mathbf{V}_1 = 0 \text{ or}$$

$$\mathbf{V}_s = -\mathbf{V}_1 - \mathbf{V}_2 = -90(1 - j) = (90 \angle 180^\circ)(1.4142 \angle -45^\circ)$$

$$\mathbf{V}_s = 127.28 \angle 135^\circ \text{ V}$$

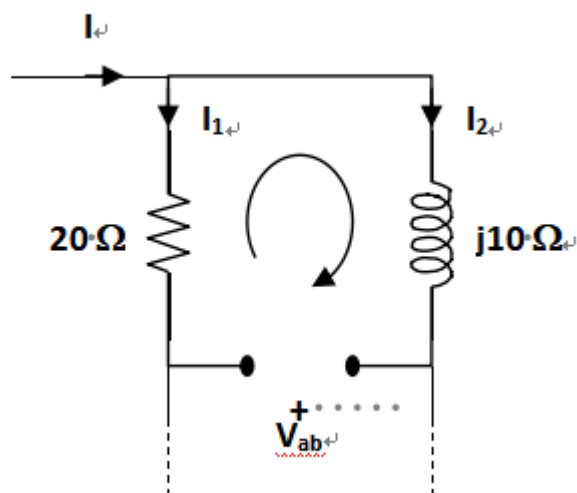
9.66 计算下图所示电路中的 $\mathbf{Z}_T$ 和 $\mathbf{V}_{ab}$



解：

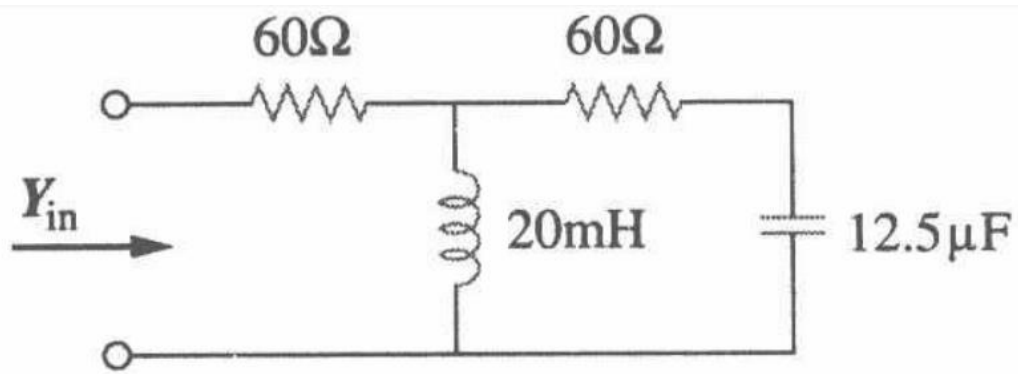
$$Z_T = (20 - j5) \parallel (40 + j10) = \frac{(20 - j5)(40 + j10)}{60 + j5} = \frac{170}{145}(12 - j)$$

$$Z_T = 14.069 - j1.172 \, \Omega = 14.118 \angle -4.76^\circ \, \Omega$$

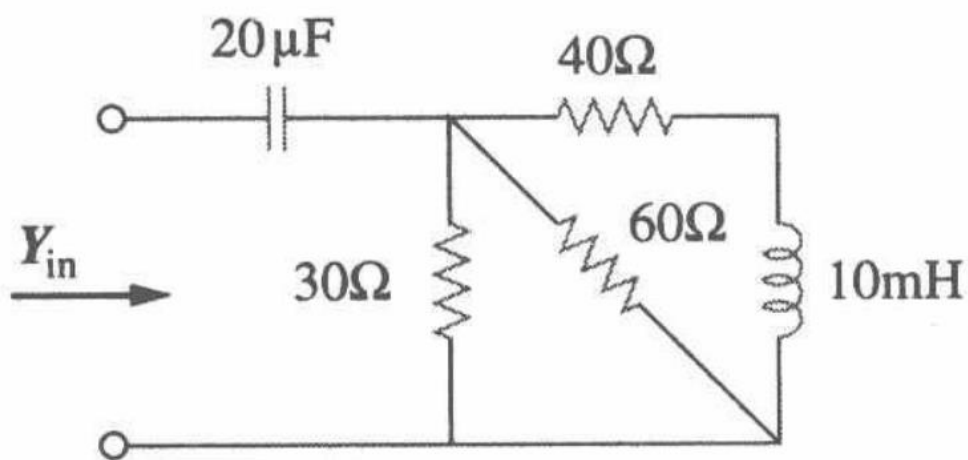


$$I = \frac{V}{Z_T} = \frac{115 \angle 90^\circ}{14.118 \angle -4.76^\circ} = 8.1456 \angle 94.76^\circ$$

9.67 计算下图所示各电路在 $\omega = 10^3 \text{ rad/s}$ 时的输入导纳。



a)



b)

解:

(a) $20 \text{ mH} \longrightarrow j\omega L = j(10^3)(20 \times 10^{-3}) = j20$

$$12.5 \text{ } \mu\text{F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(10^3)(12.5 \times 10^{-6})} = -j80$$

$$\mathbf{Z}_{\text{in}} = 60 + j20 \parallel (60 - j80)$$

$$\mathbf{Z}_{\text{in}} = 60 + \frac{(j20)(60 - j80)}{60 - j60}$$

$$\mathbf{Z}_{\text{in}} = 63.33 + j23.33 = 67.494 \angle 20.22^\circ$$

$$\mathbf{Y}_{\text{in}} = \frac{1}{\mathbf{Z}_{\text{in}}} = \mathbf{14.8 \angle -20.22^\circ \text{ mS}}$$

$$(b) \quad 10 \text{ mH} \longrightarrow j\omega L = j(10^3)(10 \times 10^{-3}) = j10$$

$$20 \mu\text{F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(10^3)(20 \times 10^{-6})} = -j50$$

$$30 \parallel 60 = 20$$

$$\mathbf{Z}_{\text{in}} = -j50 + 20 \parallel (40 + j10)$$

$$\mathbf{Z}_{\text{in}} = -j50 + \frac{(20)(40 + j10)}{60 + j10} = -j50 + 20(41.231 \angle 14.036^\circ) / (60.828 \angle 9.462^\circ)$$

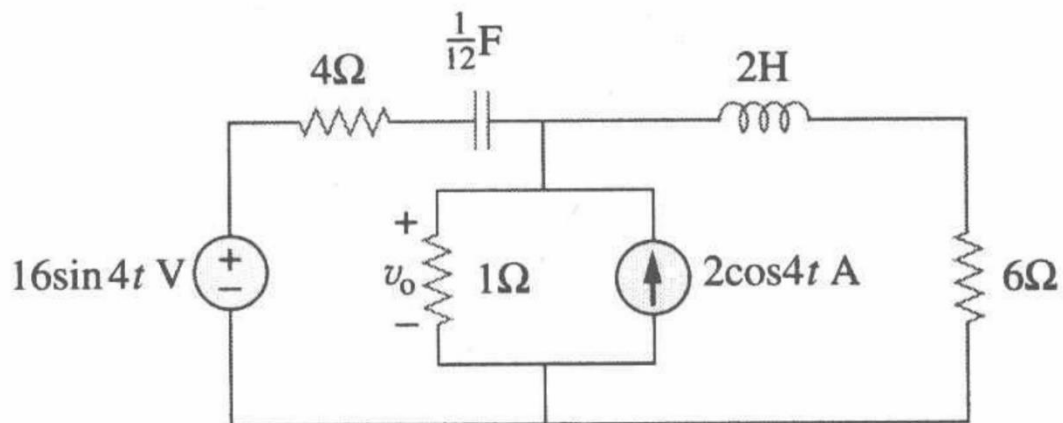
$$= -j50 + (13.5566 \angle 4.574^\circ = -j50 + 13.51342 + j1.08109$$

$$= 13.51342 - j48.9189 = 50.751 \angle -74.56^\circ$$

$$\mathbf{Z}_{\text{in}} = 13.5 - j48.92 = 50.75 \angle -74.56^\circ$$

$$\mathbf{Y}_{\text{in}} = \frac{1}{\mathbf{Z}_{\text{in}}} = \mathbf{19.704 \angle 74.56^\circ \text{ mS}} = 5.246 + j18.993 \text{ mS}$$

10.3 计算下图所示电路中的 v_o



解:

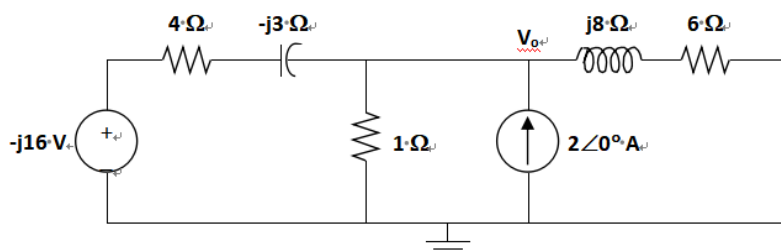
$$\omega = 4$$

$$2\cos(4t) \longrightarrow 2 \angle 0^\circ$$

$$16\sin(4t) \longrightarrow 16 \angle -90^\circ = -j16$$

$$2 \text{ H} \longrightarrow j\omega L = j8$$

$$1/12 \text{ F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(4)(1/12)} = -j3$$



运用节点分析法,

$$\frac{-j16 - V_o}{4 - j3} + 2 = \frac{V_o}{1} + \frac{V_o}{6 + j8}$$

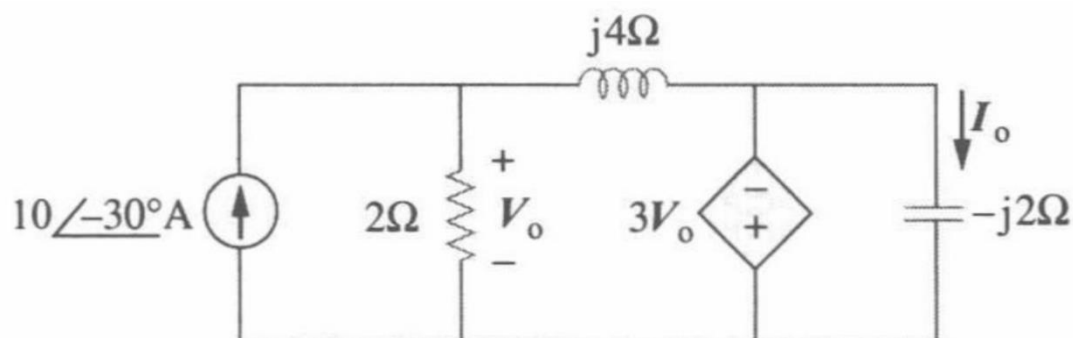
$$\frac{-j16}{4 - j3} + 2 = \left(1 + \frac{1}{4 - j3} + \frac{1}{6 + j8}\right) V_o$$

$$V_o = \frac{3.92 - j2.56}{1.22 + j0.04} = \frac{4.682 \angle -33.15^\circ}{1.2207 \angle 1.88^\circ} = 3.835 \angle -35.02^\circ$$

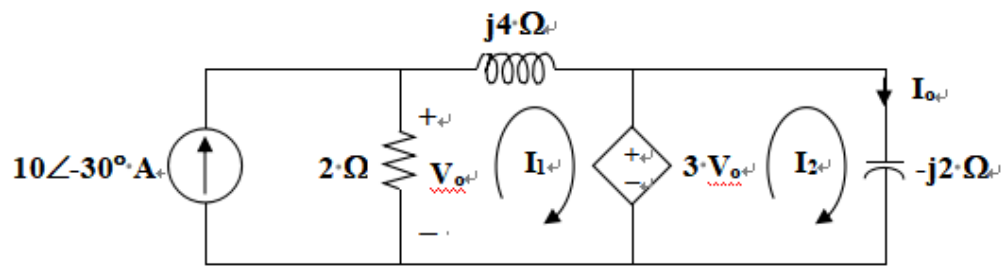
Therefore,

$$v_o(t) = 3.835 \cos(4t - 35.02^\circ) \text{ V}$$

10.32 利用网孔分析法计算下图所示电路中的 V_o 和 I_o



解:



网孔 1,

$$(2 + j4)\mathbf{I}_1 - 2(10\angle -30^\circ) + 3\mathbf{V}_o = 0$$

$$\mathbf{V}_o = 2(10\angle -30^\circ - \mathbf{I}_1)$$

因此,

$$(2 + j4)\mathbf{I}_1 - 20\angle -30^\circ + 6(10\angle -30^\circ - \mathbf{I}_1) = 0$$

$$10\angle -30^\circ = (1 - j)\mathbf{I}_1$$

$$\mathbf{I}_1 = 25\sqrt{2}\angle 15^\circ$$

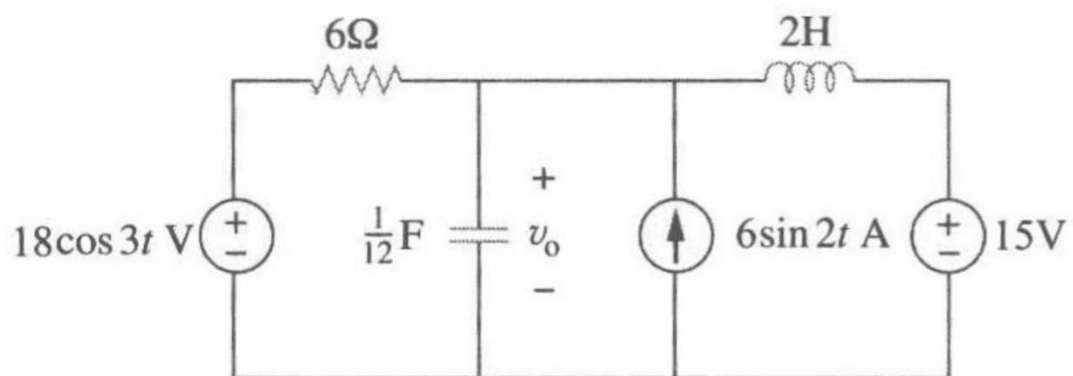
$$\mathbf{I}_o = \frac{3\mathbf{V}_o}{-j2} = \frac{3}{-j2}(2)(10\angle -30^\circ - \mathbf{I}_1)$$

$$\mathbf{I}_o = j3(10\angle -30^\circ - 5\sqrt{2}\angle 15^\circ)$$

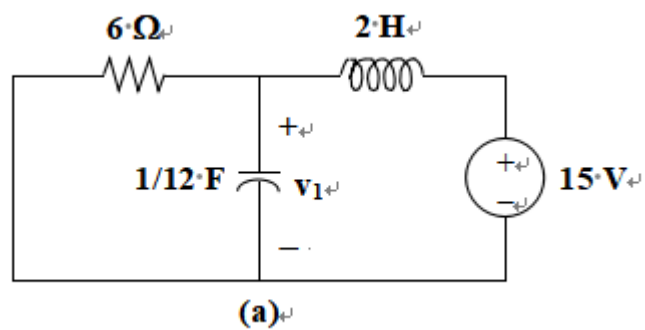
$$\mathbf{I}_o = \mathbf{21.21}\angle 15^\circ \text{ A}$$

$$\mathbf{V}_o = \frac{-j2\mathbf{I}_o}{3} = \mathbf{5.657}\angle -75^\circ \text{ V}$$

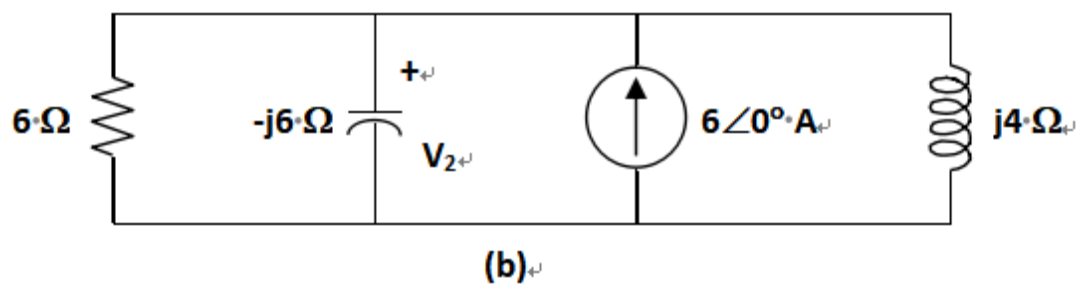
10.46 利用叠加定理计算下图所示电路中的 $v_o(t)$



解:



$$v_1 = 15V$$



$$\omega = 2$$

$$2\text{ H} \longrightarrow j\omega L = j4$$

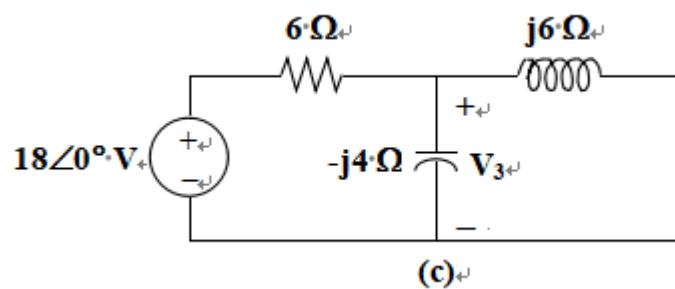
$$\frac{1}{12}\text{ F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(2)(1/12)} = -j6$$

运用节点分析法,

$$6 = \frac{\mathbf{V}_2}{6} + \frac{\mathbf{V}_2}{-j6} + \frac{\mathbf{V}_2}{j4} = \left(\frac{1}{6} + \frac{j}{6} - \frac{j}{4} \right) \mathbf{V}_2$$

$$\mathbf{V}_2 = \frac{36}{1-j0.5} = 32.18 \angle 26.57^\circ$$

因此, $v_2 = 32.18 \sin(2t + 26.57^\circ) \text{ V}$



$$\omega = 3$$

$$2 \text{ H} \longrightarrow j\omega L = j6$$

$$\frac{1}{12} \text{ F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(3)(1/12)} = -j4$$

$$\frac{18 - \mathbf{V}_3}{6} = \frac{\mathbf{V}_3}{-j4} + \frac{\mathbf{V}_3}{j6}$$

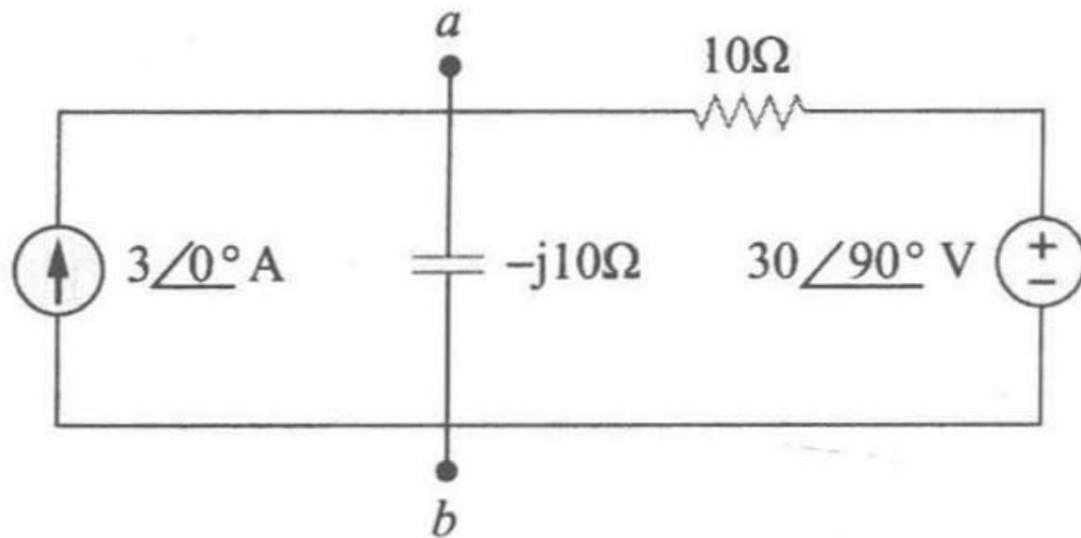
$$\mathbf{V}_3 = \frac{18}{1+j0.5} = 16.1 \angle -26.57^\circ$$

因此, $v_3 = 16.1 \cos(3t - 26.57^\circ) \text{ V}$

最后,

$$v_o(t) = [15 + 32.18 \sin(2t + 26.57^\circ) + 16.1 \cos(3t - 26.57^\circ)] \text{ V}$$

10.58 求下图所示各电路在端口 a-b 处的诺顿等效电路。



解:

$$V_{oc} = V_{ab} \quad -3 + [(V_{ab}-0)/(-j10)] + [(V_{ab}-j30)/10] = 0$$

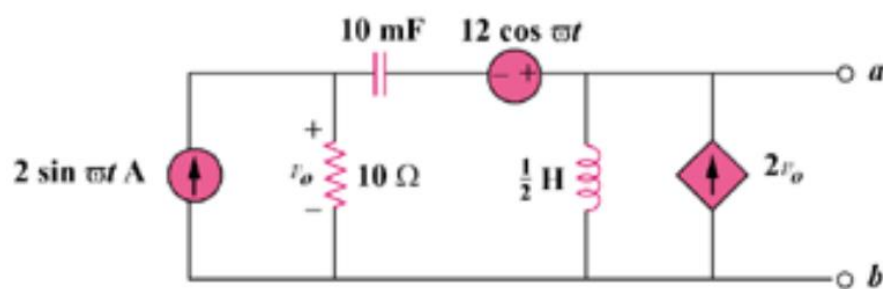
$$(0.1+j0.1)V_{ab} = 3+j3$$

$$V_{oc} = V_{Thev} = 3(1+j)/[0.1(1+j)] = 30 \text{ V.}$$

$$I_{sc} = 3 + [(j30)/10] = 3+j3.$$

$$Z_{eq} = V_{Thev}/I_{sc} = 30/[3(1+j)] = (5-j5) \Omega.$$

10.66 求下图所示电路在端口 a-b 处的戴维南等效电路与诺顿等效电路，假设 $\omega = 10 \text{ rad/s}$



解:

$$\omega = 10$$

$$0.5 \text{ H} \longrightarrow j\omega L = j(10)(0.5) = j5$$

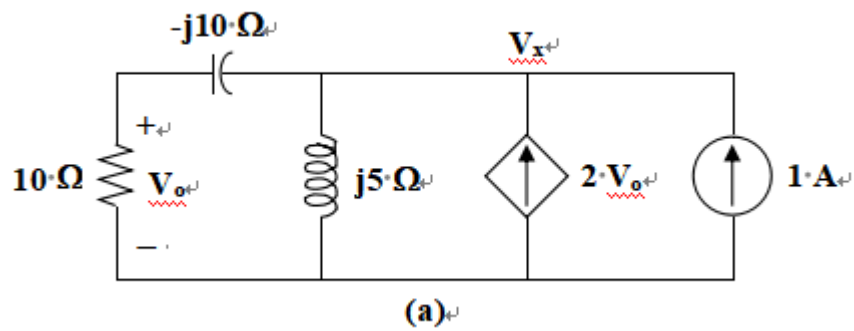
$$10 \text{ mF} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(10)(10 \times 10^{-3})} = -j10$$

To

find

Z_{th}

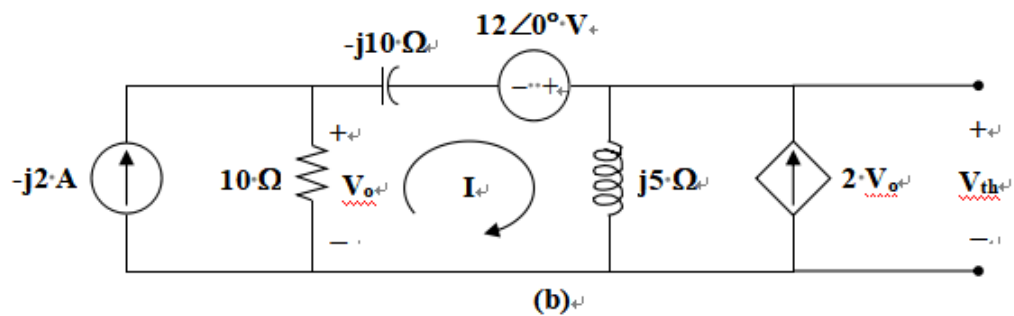
,



$$1 + 2V_o = \frac{V_x}{j5} + \frac{V_x}{10 - j10}, \quad V_o = \frac{10V_x}{10 - j10}$$

$$1 + \frac{19V_x}{10 - j10} = \frac{V_x}{j5} \longrightarrow V_x = \frac{-10 + j10}{21 + j2}$$

$$Z_N = Z_{th} = \frac{V_x}{1} = \frac{14.142 \angle 135^\circ}{21.095 \angle 5.44^\circ} = 670 \angle 129.56^\circ \text{ m}\Omega$$



$$(10 - j10 + j5)I - (10)(-j2) + j5(2V_o) - 12 = 0$$

$$V_o = (10)(-j2 - I)$$

因此,

$$(10 - j105)I = -188 - j20$$

$$I = \frac{188 + j20}{-10 + j105}$$

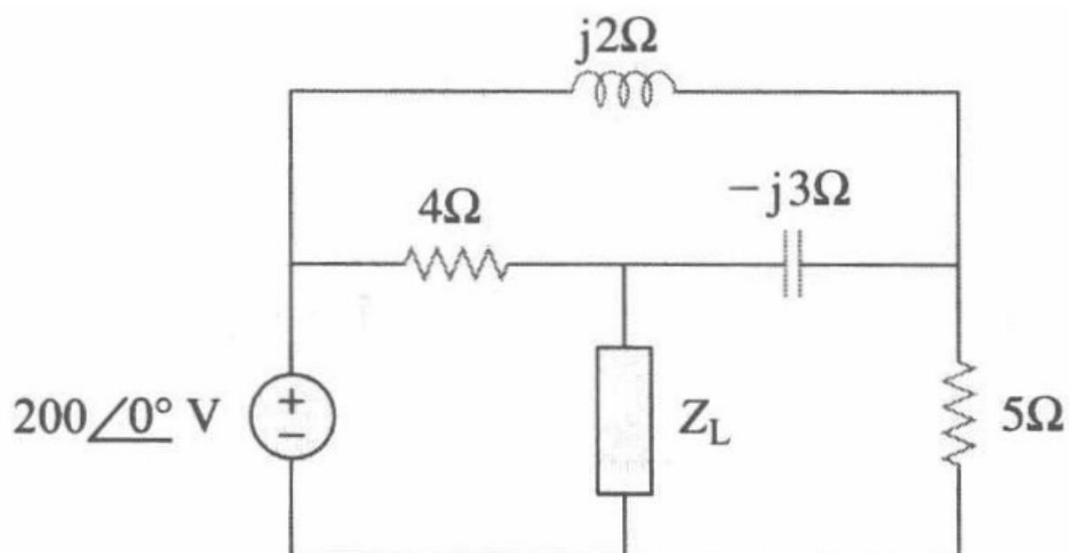
$$V_{th} = j5(I + 2V_o) = j5(-19I - j40) = -j95I + 200$$

$$\begin{aligned}\mathbf{V}_{th} &= \frac{-j95(188 + j20)}{-10 + j105} + 200 = \frac{(95\angle -90^\circ)(189.06\angle 6.07^\circ)}{105.48\angle 95.44} + 200 \\ &= 170.28\angle -179.37^\circ + 200 = -170.27 - j1.8723 + 200 = 29.73 - j1.8723\end{aligned}$$

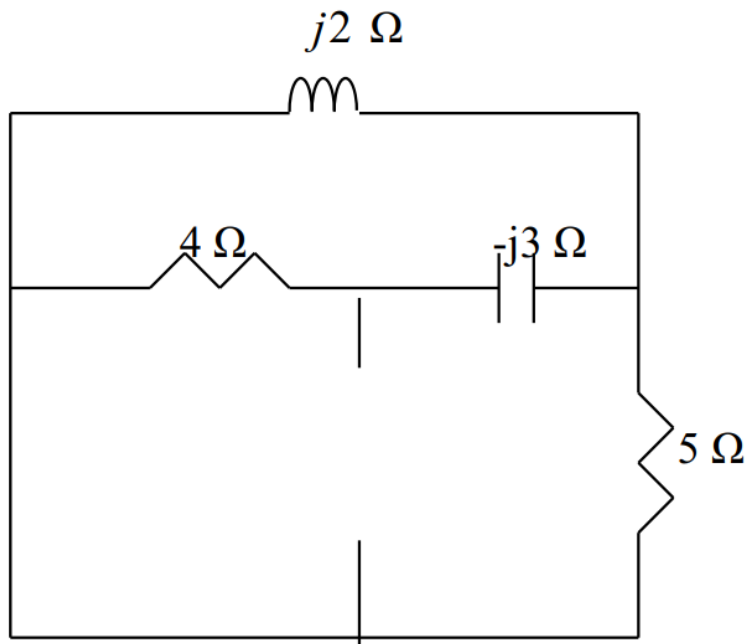
$$\mathbf{V}_{th} = 29.79\angle -3.6^\circ \text{ V}$$

$$\mathbf{I}_N = \frac{\mathbf{V}_{th}}{\mathbf{Z}_{th}} = \frac{29.79\angle -3.6^\circ}{0.67\angle 129.56^\circ} = 44.46\angle -133.16^\circ \text{ A}$$

11.12 对于下图所示电路，试确定实现最大功率传输（对于 Z_L ）时的负载阻抗 Z_L ，并计算负载吸收的最大功率值。



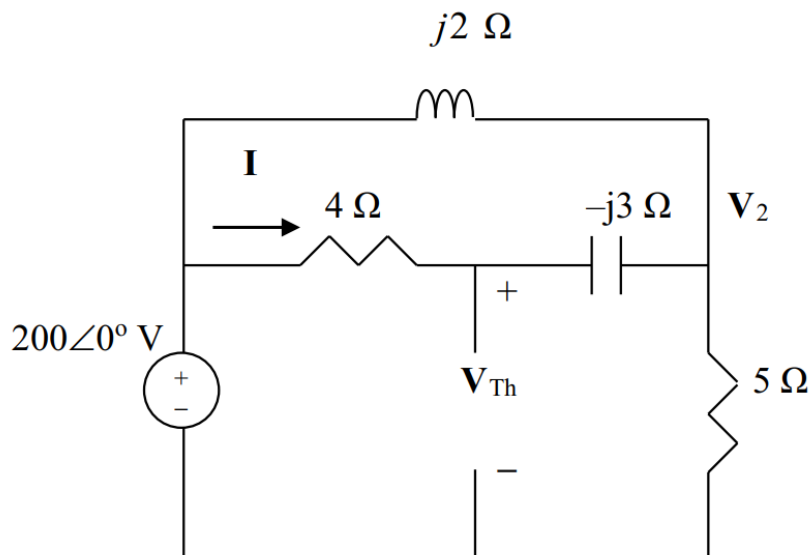
解：



$$Z_{\text{Thev}} = \frac{4 \left(-j3 + \frac{5 \times j2}{5 + j2} \right)}{4 - j3 + \frac{5 \times j2}{5 + j2}} = \frac{4(0.6896 - j1.2758)}{4.69 - j1.2758} = \frac{4(1.4502 \angle -61.61^\circ)}{4.86 \angle -15.22^\circ}$$

$$= 1.1936 \angle -46.39^\circ$$

$$Z_{\text{Thev}} = 0.8233 - j0.8642 \quad \text{or} \quad Z_L = [823.3 + j864.2] \text{ m}\Omega.$$



$$\frac{V_2 - 200}{4 - j3} + \frac{V_2 - 200}{j2} + \frac{V_2 - 0}{5} = 0$$

$$(0.16 + j0.12 - j0.5 + 0.2)V_2 = (0.16 + j0.12 - j0.5)200$$

$$(0.5235 \angle -46.55^\circ)V_2 = (0.4123 \angle -67.17^\circ)200 = 82.46 \angle -67.17^\circ$$

Thus, $\mathbf{V}_2 = 157.52\angle -20.62^\circ \text{ V} = 147.43 - j55.473$

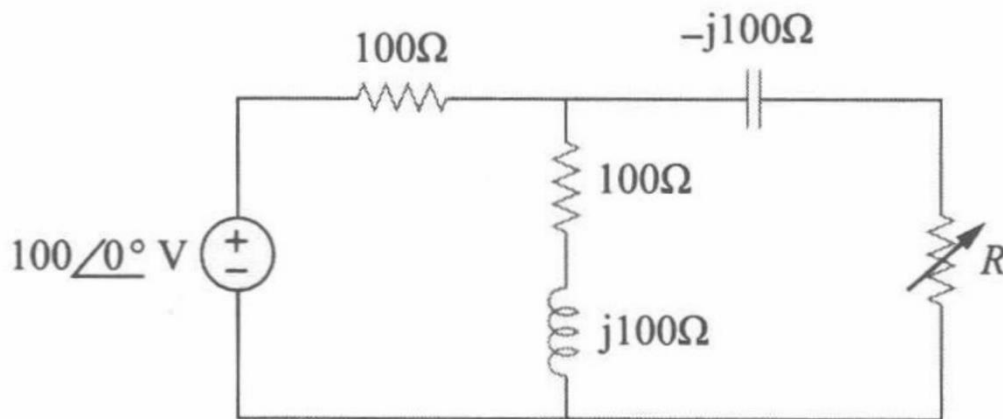
$$\begin{aligned} \mathbf{I} &= (200 - \mathbf{V}_2)/(4 - j3) = (200 - 147.43 + j55.473)/(4 - j3) \\ &= (52.57 + j55.473)/(4 - j3) = (76.426\angle 46.54^\circ)/(5\angle -36.87^\circ) \\ &= 15.285\angle 83.41^\circ = 1.7542 + j15.184 \end{aligned}$$

$$\begin{aligned} \mathbf{V}_{\text{Thev}} &= 200 - 4\mathbf{I} = 200 - 7.0168 - j60.736 = [192.983 - j60.736] \text{ V} \\ &= 202.31\angle -17.47^\circ \text{ V} \end{aligned}$$

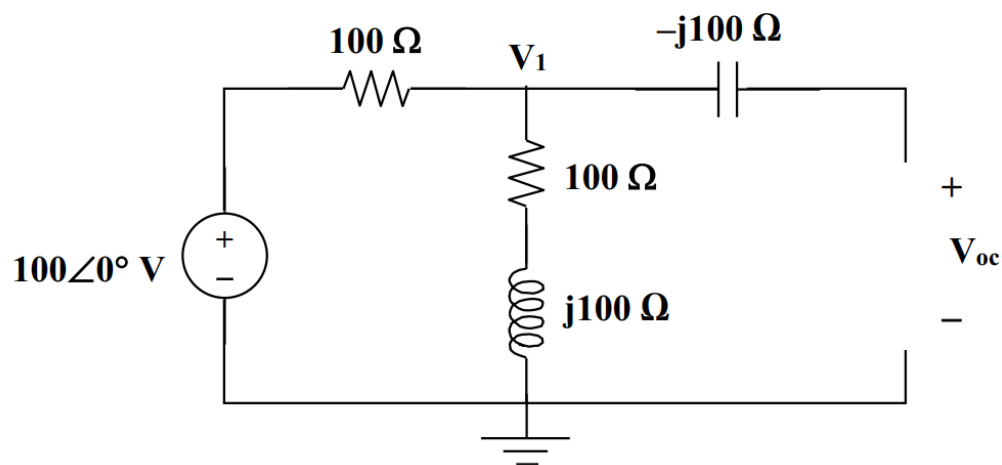
$$|\mathbf{I}_L| = (202.31/(2 \times 0.8233)) = 122.865 \text{ A}$$

$$P_{\text{avg}} = [(|\mathbf{I}_L|)^2 \times 0.8233]/2 = \mathbf{6.214 \text{ kW.}}$$

11.19 调节下图所示电路中的可变电阻 R 使其吸收最大的平均功率，试求该电阻值以及所吸收的最大平均功率。



解：



$$\mathbf{Z}_{\text{eq}} = -j100 + 100(100 + j100)/(100 + 100 + j100).$$

$$[(\mathbf{V}_1 - 100)/100] + [(\mathbf{V}_1 - 0)/(100 + j100)] + 0 = 0, \quad \mathbf{V}_{\text{oc}} = \mathbf{V}_{\text{Thev}} = \mathbf{V}_1.$$

$$\begin{aligned} \mathbf{Z}_{\text{eq}} &= -j100 + 100(1.4142\angle 45^\circ)/(2.2361\angle 26.57^\circ) = -j100 + 63.244\angle 18.43^\circ \\ &= -j100 + 60 + j20 = (60 - j80) \Omega = 100\angle -53.13^\circ \Omega. \end{aligned}$$

$$(0.01 + 0.005 - j0.005)\mathbf{V}_1 = 1 = 0.0158114\angle -18.43^\circ \mathbf{V}_1, \quad \mathbf{V}_1 = 63.246\angle 18.43^\circ.$$

因此

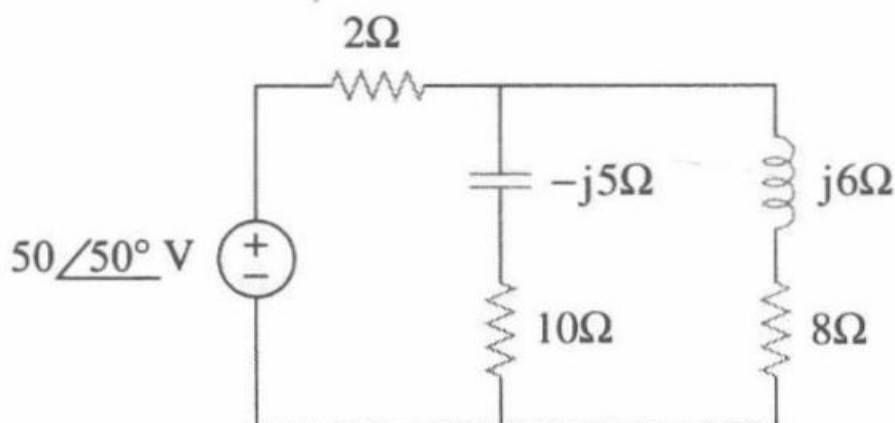
$$R = 100 \Omega$$

$$|\mathbf{I}| = 63.246/|60 - j80 + 100| = 63.246/178.885 = 0.353557 \text{ A}$$

$$P_{\text{avg}} = [(0.353557)^2/2]100 = 6.25 \text{ W}.$$

11.51 对于下图所示的电路，试计算：（a）功率因数；（b）电源传递的平均功率；（c）无功功率；

（d）视在功率；（e）复功率。



解：

$$(a) \quad \mathbf{Z}_T = 2 + (10 - j5) \parallel (8 + j6)$$

$$\mathbf{Z}_T = 2 + \frac{(10 - j5)(8 + j6)}{18 + j} = 2 + \frac{110 + j20}{18 + j}$$

$$\mathbf{Z}_T = 8.152 + j0.768 = 8.188\angle 5.382^\circ$$

$$\text{pf} = \cos(5.382^\circ) = 0.9956 \quad (\text{lagging})$$

$$(b) \quad \mathbf{S} = \mathbf{V}\mathbf{I}^* = \frac{|\mathbf{V}|^2}{(\mathbf{Z}_T)^*} = \frac{(50)^2}{(8.188\angle -5.382^\circ)}$$

$$\mathbf{S} = 305.325\angle 5.382^\circ$$

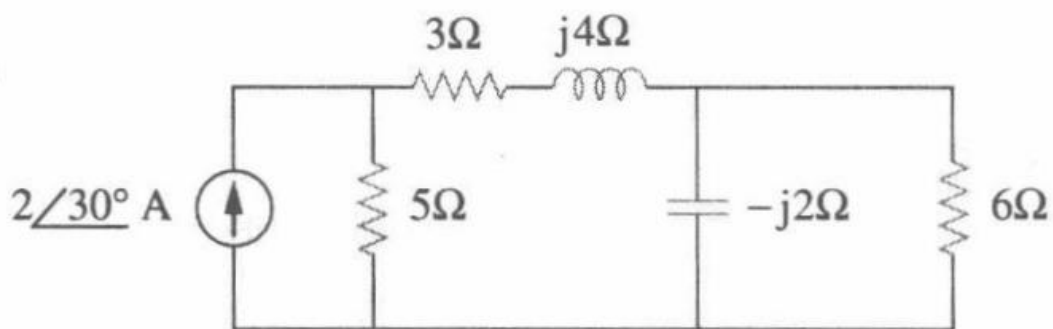
$$P = S \cos \theta = \mathbf{304\ W}$$

$$(c) \quad Q = S \sin \theta = \mathbf{28.64\ VAR}$$

$$(d) \quad S = |\mathbf{S}| = \mathbf{305.3\ VA}$$

$$(e) \quad \mathbf{S} = 305.325 \angle 5.382^\circ = \mathbf{(304 + j28.64)\ VA}$$

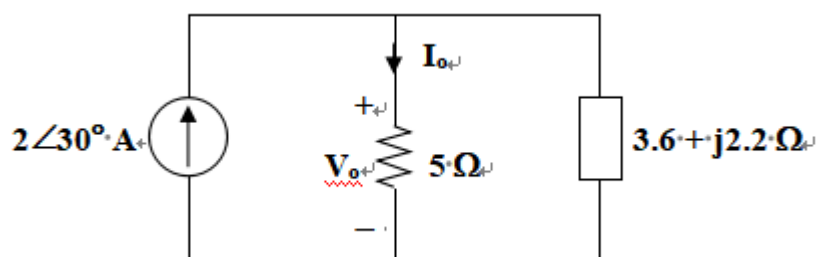
11.56 确定下图所示电路中的电源传递的复功率。



解：

$$-j2 \parallel 6 = \frac{(6)(-j2)}{6 - j2} = \frac{12 \angle -90^\circ}{6.32456 \angle -18.435^\circ} = 1.897365 \angle -71.565^\circ = 0.6 - j1.8$$

$$3 + j4 + [(-j2) \parallel 6] = 3.6 + j2.2$$



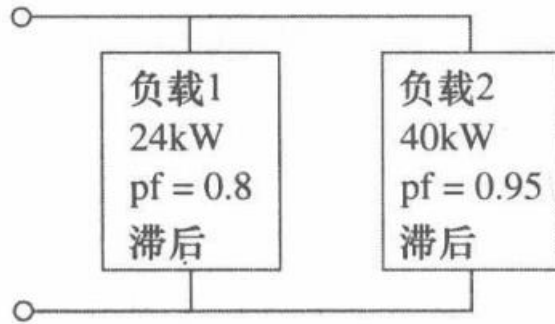
$$\mathbf{I_o} = \frac{3.6 + j2.2}{8.6 + j2.2} (2 \angle 30^\circ) = \frac{4.219 \angle 31.4296^\circ}{8.87694 \angle 14.3493^\circ} (2 \angle 30^\circ) = 0.95055 \angle 47.08^\circ$$

$$\mathbf{V_o} = 5 \mathbf{I_o} = 4.75275 \angle 47.08^\circ$$

$$\mathbf{S} = \mathbf{V}_o \mathbf{I}_s^* = (4.75275 \angle 47.08^\circ)(2 \angle -30^\circ)$$

$$\mathbf{S} = 9.5055 \angle 17.08^\circ = (9.086 + j2.792) \text{ VA}$$

11.74 某 120Vrms、60Hz 电源给两个相互并联的负载供电，如下图所示。(a) 试求该并联负载的功率因数；(b) 试计算将功率因数提高到 1，所需并联的电容值。



解：

$$(a) \quad \theta_1 = \cos^{-1}(0.8) = 36.87^\circ$$

$$S_1 = \frac{P_1}{\cos \theta_1} = \frac{24}{0.8} = 30 \text{ kVA}$$

$$Q_1 = S_1 \sin \theta_1 = (30)(0.6) = 18 \text{ kVAR}$$

$$\mathbf{S}_1 = 24 + j18 \text{ kVA}$$

$$\theta_2 = \cos^{-1}(0.95) = 18.19^\circ$$

$$S_2 = \frac{P_2}{\cos \theta_2} = \frac{40}{0.95} = 42.105 \text{ kVA}$$

$$Q_2 = S_2 \sin \theta_2 = 13.144 \text{ kVAR}$$

$$\mathbf{S}_2 = 40 + j13.144 \text{ kVA}$$

$$\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2 = 64 + j31.144 \text{ kVA}$$

$$\theta = \tan^{-1}\left(\frac{31.144}{64}\right) = 25.95^\circ$$

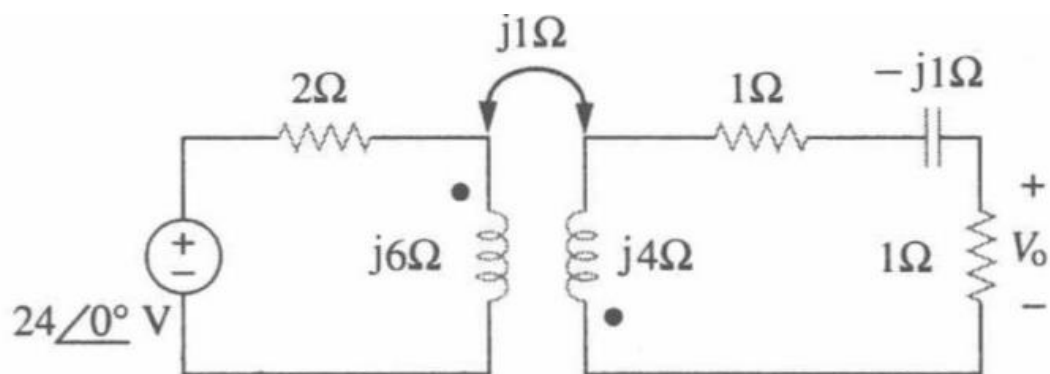
$$\text{pf} = \cos \theta = \mathbf{0.8992}$$

(b) $\theta_2 = 25.95^\circ$, $\theta_1 = 0^\circ$

$$Q_c = P[\tan \theta_2 - \tan \theta_1] = 64[\tan(25.95^\circ) - 0] = 31.144 \text{ kVAR}$$

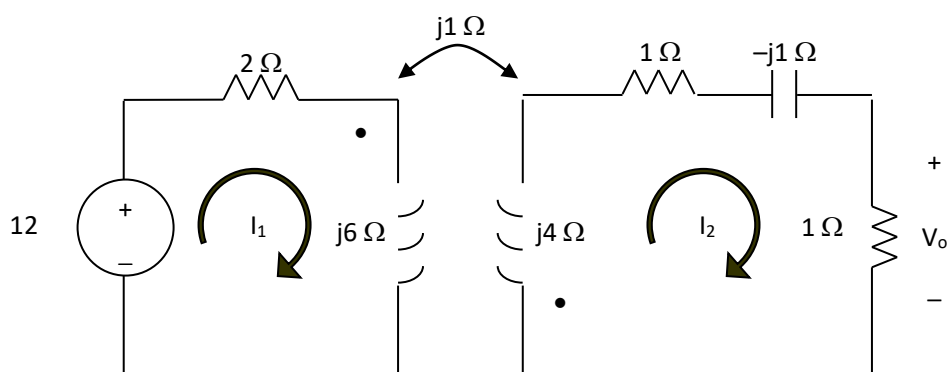
$$C = \frac{Q_c}{\omega V_{\text{rms}}^2} = \frac{31,144}{(2\pi)(60)(120)^2} = 5.74 \text{ mF}$$

13.7 计算下图所示电路中的 V_o



解：

运用网孔分析法：



对于网孔 1,

$$(2+j6)i_1 + j i_2 = 24$$

对于网孔 2,

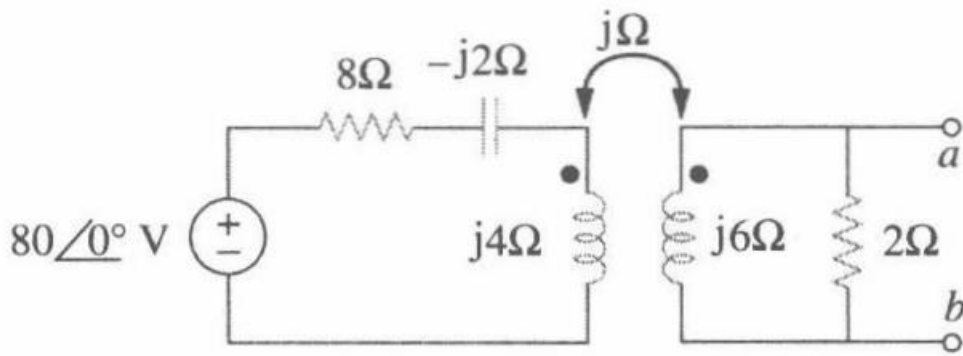
$$j i_1 + (2-j+j4)i_2 = j i_1 + (2+j3)i_2 = 0 \text{ or } i_1 = (-3+j2)i_2$$

解得

$$i_2 = (-0.8762+j0.6328) \text{ A.}$$

$$V_o = I_2 \times 1 = 1.081 \angle 144.16^\circ \text{ V.}$$

13.11 利用网孔分析法求解下图所示电路中的 i_x 其中: $i_s = 4\cos 600t \text{ A}$, $v_s = 110\cos(600t + 30^\circ) \text{ V}$



解:

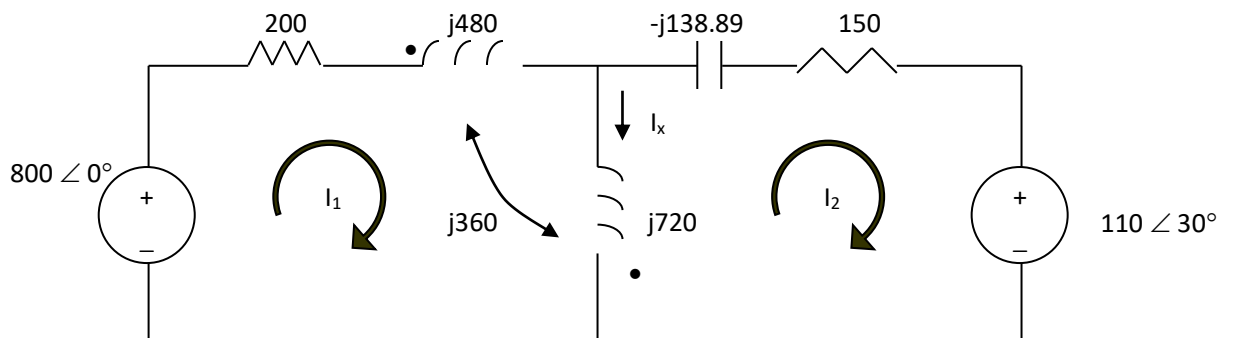
$$800\text{mH} \longrightarrow j\omega L = j600 \times 800 \times 10^{-3} = j480$$

$$600\text{mH} \longrightarrow j\omega L = j600 \times 600 \times 10^{-3} = j360$$

$$1200\text{mH} \longrightarrow j\omega L = j600 \times 1200 \times 10^{-3} = j720$$

$$12\mu\text{F} \rightarrow \frac{1}{j\omega C} = \frac{-j}{600 \times 12 \times 10^{-6}} = -j138.89$$

把电流源变换为电压源,



对于网孔 1,

$$800 = (200 + j480 + j720)I_1 + j360I_2 - j720I_2 \text{ or}$$

$$800 = (200 + j1200)I_1 - j360I_2$$

对于网孔 2,

$$110\angle 30^\circ + 150 - j138.89 + j720 I_2 + j360 I_1 = 0 \text{ or}$$

$$-95.2628 - j55 = -j360 I_1 + (150 + j581.1) I_2$$

写成矩阵形式

$$\begin{bmatrix} 800 \\ -95.2628 - j55 \end{bmatrix} = \begin{bmatrix} 200 + j1200 & -j360 \\ -j360 & 150 + j581.1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

解得

$$I =$$

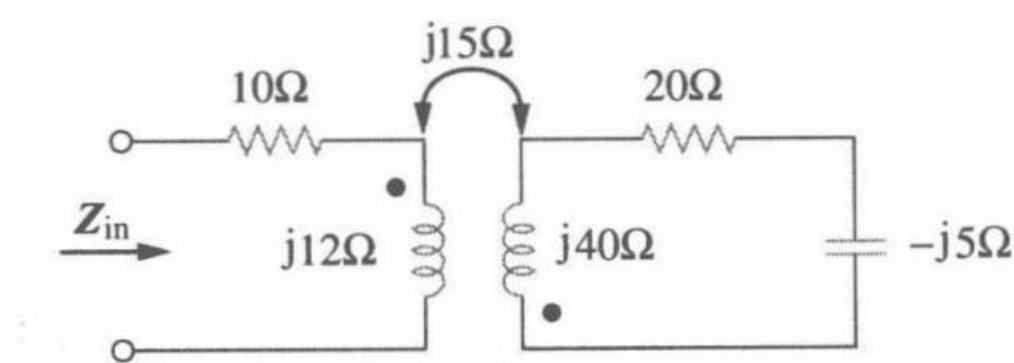
$$0.1390 - 0.7242i$$

$$0.0609 - 0.2690i$$

$$I_x = I_1 - I_2 = 0.0781 - j0.4552 = 0.4619\angle -80.26^\circ$$

$$\text{因此, } i_x(t) = 461.9\cos(600t - 80.26^\circ) \text{ mA.}$$

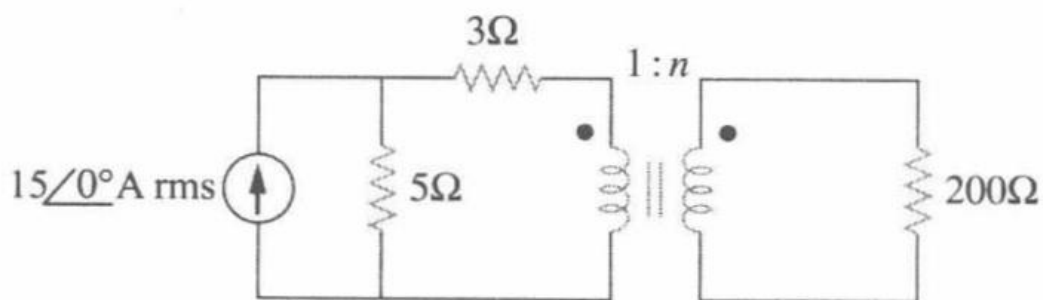
13.33 计算下图所示空心变压器电路的输入阻抗。



解:

$$\begin{aligned} Z_{in} &= 10 + j12 + (15)^2 / (20 + j40 - j5) = 10 + j12 + 225 / (20 + j35) \\ &= 10 + j12 + 225(20 - j35) / (400 + 1225) \\ &= (12.769 + j7.154) \Omega \end{aligned}$$

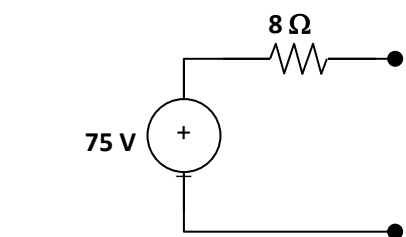
13.53 对于下图所示网络:(a)试求传递给 200Ω 负载功率最大时的匝数比 n ;(b)如果 $n=10$, 计算 200Ω 负载的功率。



解:

(a)

变压器左边的戴维南等效电路为:



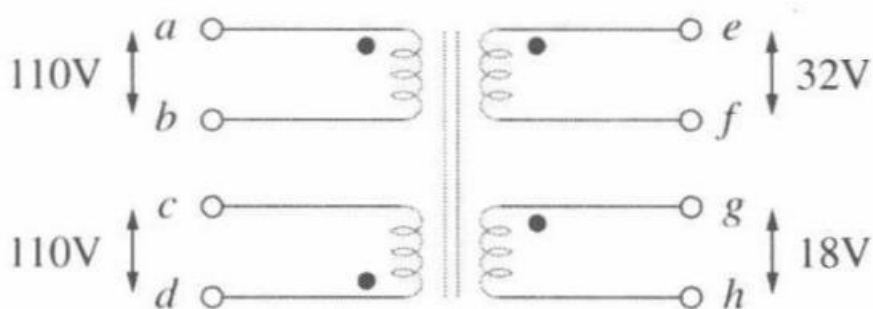
$$Z_L' = Z_L/n^2 = 200/n^2$$

$$8 = 200/n^2, \quad n = 5.$$

(b) If $n = 10$, $Z_L' = 200/100 = 2 \Omega$ and $I = 75/(8+2) = 7.5 \text{ A}$

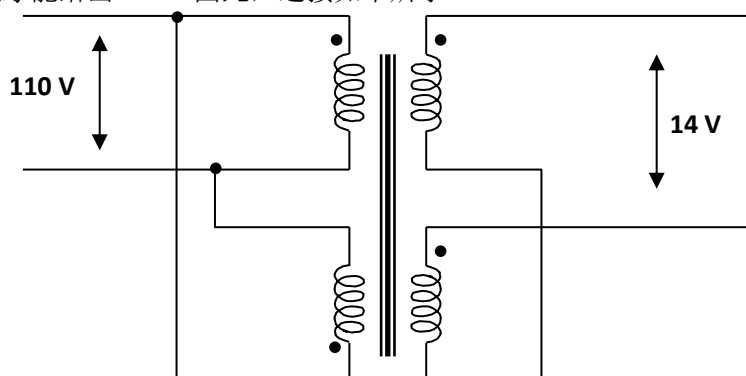
$$p = I^2 Z_L' = (7.5)^2(2) = 112.5 \text{ W}.$$

13.93 下图所示的四绕组变压器通常用在既可以在 110V 电压下工作又可以在 220V 电压下工作的设备（如计算机、录像机等）中，这就使得这类设备既可以在国内使用，也可以在国外使用，试说明提供如下电压所需的变压器连接方式：（a）输入 1100V 时，输出 14V；（b）输入 220V 时，输出 50V。

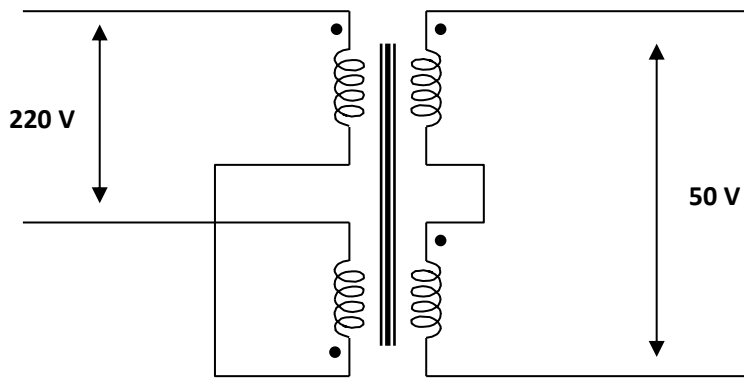


解:

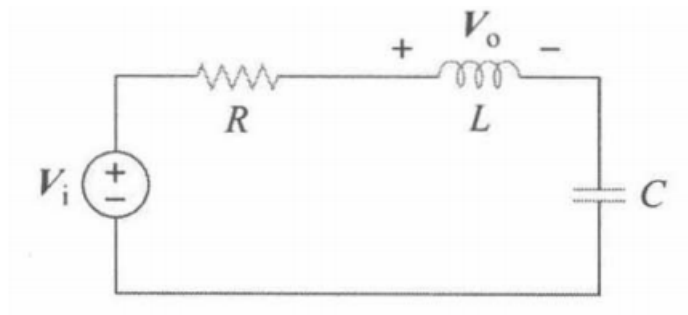
（a）对于 110V 的输入，初级绕组必须平行连接，次级上有串联辅助。线圈必须是相反的串联才能给出 14V。因此，连接如下所示。



（b）为了在一次侧得到 220V，线圈是串联串联辅助在二次侧。线圈必须连接串联，以提供 50V。因此，连接如下所示

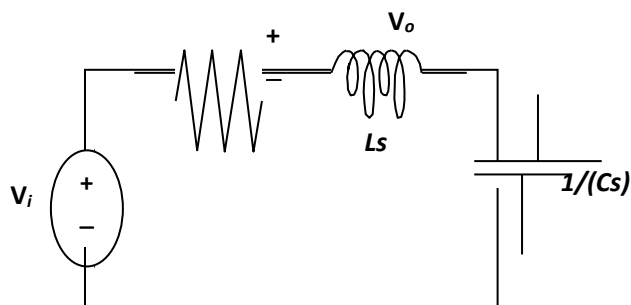


14.4 求下图所示电路的传递函数 $H(s) = V_o/V_i$



解：

把电路转为到 s 域



$$-V_i + RI + LsI + [1/(Cs)]I = 0, \quad V_o = LsI$$

$$[R + Ls + 1/(Cs)]I = V_i \text{ or } I = [Cs / (CLs^2 + CRs + 1)]V_i.$$

$$V_o = LsI,$$

$$H(s) = V_o/V_i = LCs^2 / (CLs^2 + CRs + 1).$$

14.30 电路由电感值为 10mH、电阻值为 20Ω 的线圈，电容器和电压均值为 120V 的信号发生器串联组成。试求：(a) 使电路在 15kHz 时发生谐振的电容值；(b) 谐振时通过线圈的电流；(c) 电路的 Q 值。

解：

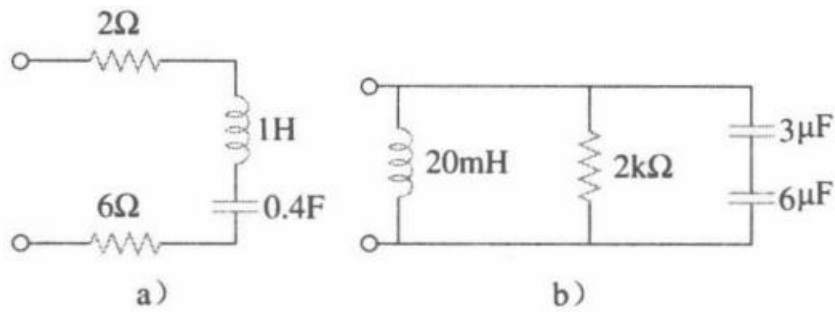
$$(a) \quad f_o = 15,000 \text{ Hz leads to } \omega_o = 2\pi f_o = 94.25 \text{ krad/s} = 1/(LC)^{0.5}$$

$$LC = 1/8.883 \times 10^9 \text{ or } C = 1/(8.883 \times 10^9 \times 10^{-2}) = 11.257 \times 10^{-9} \text{ F} = \mathbf{11.257 \text{ pF}}.$$

$$(b) \quad I = 120/20 = \mathbf{6 \text{ A}}.$$

(c) $Q = \omega_0 L / R = 94.25 \times 10^3 (0.01) / 20 = \mathbf{47.12}$.

14.42 对于下图所示电路，求谐振频率 ω_0 、品质因数 Q 及带宽 B 。



解：

(a)

$$R = 2 + 6 = 8\Omega, \quad L = 1\text{ H}, \quad C = 0.4\text{ F}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.4}} = \mathbf{1.5811\text{ rad/s}}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1.5811}{8} = \mathbf{0.1976}$$

$$B = \frac{R}{L} = \mathbf{8\text{ rad/s}}$$

(b)

$$3\text{ }\mu\text{F and } 6\text{ }\mu\text{F} \longrightarrow \frac{(3)(6)}{3+6} = 2\text{ }\mu\text{F}$$

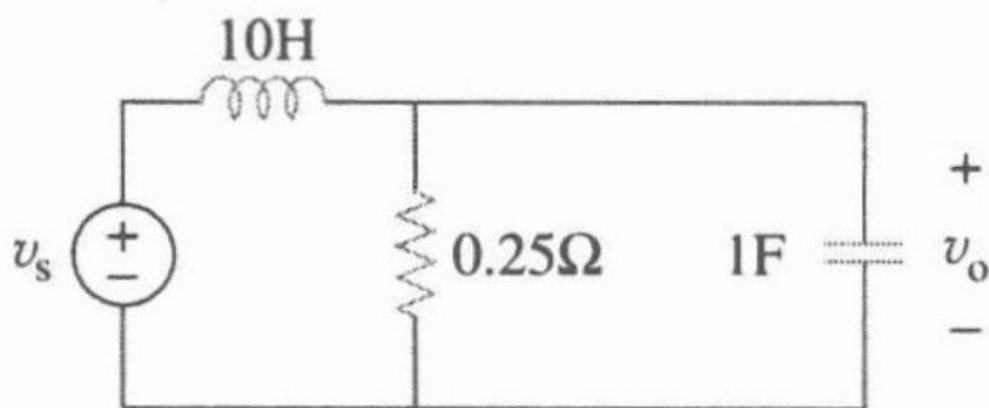
$$C = 2\text{ }\mu\text{F}, \quad R = 2\text{ k}\Omega, \quad L = 20\text{ mH}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(2 \times 10^{-6})(20 \times 10^{-3})}} = \mathbf{5\text{ krad/s}}$$

$$Q = \frac{R}{\omega_0 L} = \frac{2 \times 10^3}{(5 \times 10^3)(20 \times 10^{-3})} = 20$$

$$B = \frac{1}{RC} = \frac{1}{(2 \times 10^3)(2 \times 10^{-6})} = 250 \text{ rad/s}$$

14.48 求下图所示电路的传递函数 V_o/V_s ，并证明该电路为低通滤波器



解：

$$\mathbf{H}(\omega) = \frac{R \parallel \frac{1}{j\omega C}}{j\omega L + R \parallel \frac{1}{j\omega C}}$$

$$\mathbf{H}(\omega) = \frac{\frac{R/j\omega C}{R + 1/j\omega C}}{j\omega L + \frac{R/j\omega C}{R + 1/j\omega C}} = \frac{\frac{R}{1 + j\omega RC}}{j\omega L + \frac{R}{1 + j\omega RC}} = \frac{R}{R + j\omega L + \frac{R/j\omega C}{R + 1/j\omega C}}$$

$$\mathbf{H}(\omega) = \frac{0.25}{(0.25 - \omega^2 2.5) + j\omega 10}$$

$H(0) = 1$ and $H(\infty) = 0$ 所以是低通滤波器.

14.55 确定 $R=10\Omega$, $L=25\text{mH}$, $C=0.4\mu\text{F}$ 的 RLC 串联带通滤波器的频率范围, 并计算其品质因数。

解:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(25 \times 10^{-3})(0.4 \times 10^{-6})}} = 10 \text{ krad/s}$$

$$B = \frac{R}{L} = \frac{10}{25 \times 10^{-3}} = 0.4 \text{ krad/s}$$

$$Q = \frac{10}{0.4} = 25$$

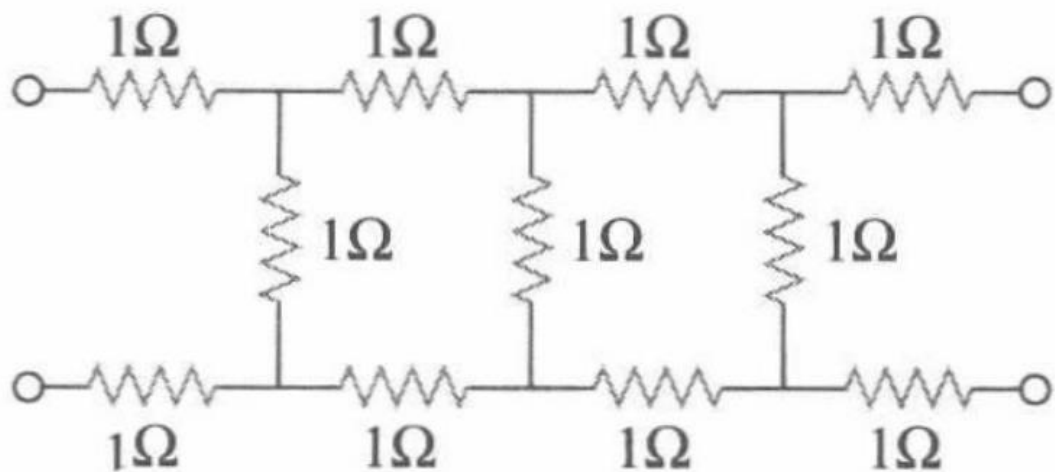
$$\omega_1 = \omega_0 - B/2 = 10 - 0.2 = 9.8 \text{ krad/s} \quad \text{or} \quad f_1 = \frac{9.8}{2\pi} = 1.56 \text{ kHz}$$

$$\omega_2 = \omega_0 + B/2 = 10 + 0.2 = 10.2 \text{ krad/s} \quad \text{or} \quad f_2 = \frac{10.2}{2\pi} = 1.62 \text{ kHz}$$

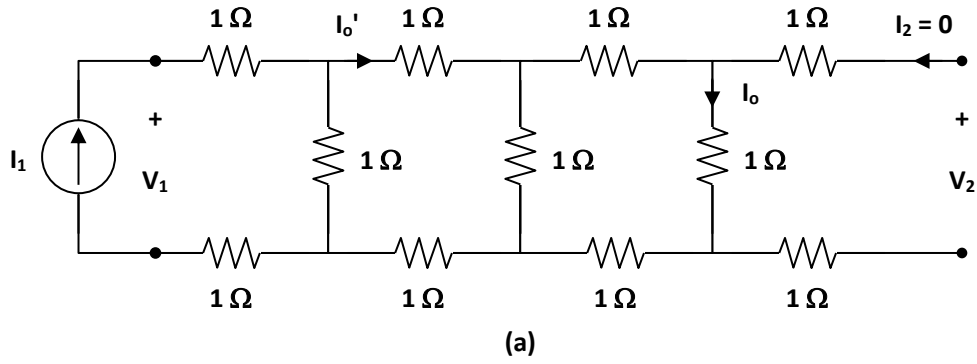
因此,

$$1.56 \text{ kHz} < f < 1.62 \text{ kHz}$$

19.2 求下图所示网络的等效阻抗参数。



解:



$$z_{11} = \frac{V_1}{I_1} = 2 + 1 \parallel [2 + 1 \parallel (2 + 1)]$$

$$z_{11} = 2 + 1 \parallel \left(2 + \frac{3}{4}\right) = 2 + \frac{(1)(11/4)}{1 + 11/4} = 2 + \frac{11}{15} = 2.733$$

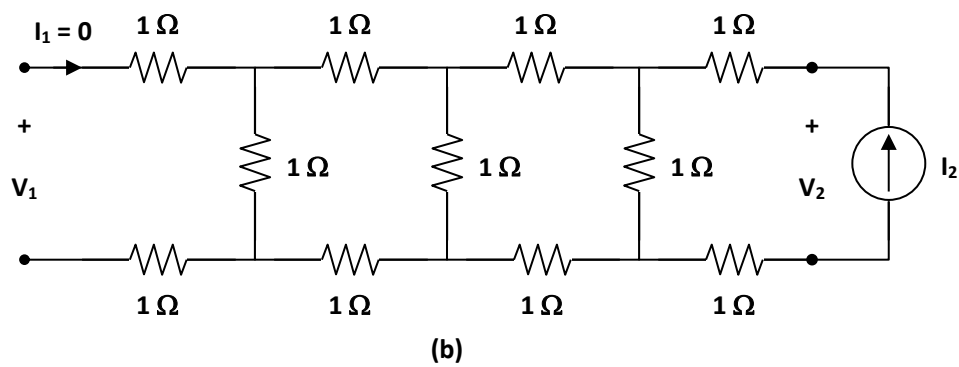
$$I_o = \frac{1}{1+3} I_o' = \frac{1}{4} I_o'$$

$$I_o' = \frac{1}{1+11/4} I_1 = \frac{4}{15} I_1$$

$$I_o = \frac{1}{4} \cdot \frac{4}{15} I_1 = \frac{1}{15} I_1$$

$$V_2 = I_o = \frac{1}{15} I_1$$

$$z_{21} = \frac{V_2}{I_1} = \frac{1}{15} = z_{12} = 0.06667$$



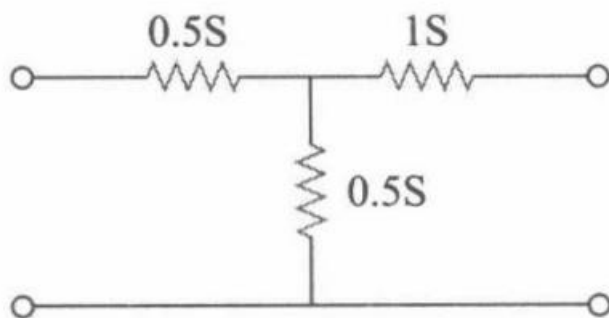
$$z_{22} = \frac{V_2}{I_2} = 2 + 1 \parallel (2 + 1 \parallel 3) = z_{11} = 2.733$$

因此,

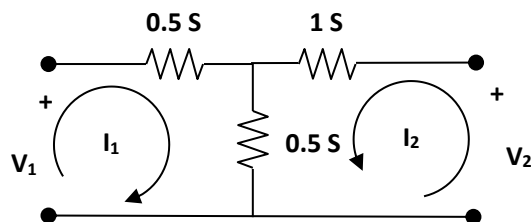
$$[\mathbf{z}] = \begin{bmatrix} 2.733 & 0.06667 \\ 0.06667 & 2.733 \end{bmatrix} \Omega$$

$$\mathbf{z}_{11} = \frac{\mathbf{V}_1}{\mathbf{I}_1} = 2 + 1 \parallel [2 + 1 \parallel (2 + 1)]$$

19.18 求下图所示二端口电路的 $\mathbf{y}$ 参数



解:



$$I_1 = y_{11}V_1 + y_{12}V_2$$

$$I_2 = y_{21}V_1 + y_{22}V_2$$

$$\text{令 } V_1 = 1 \text{ V}, \quad V_2 = 0, \quad V_1 = 0, \quad V_2 = 1 \text{ V.}$$

$$I_1 = 1 / [(1/0.5) + (1/1.5)] = 1 / (2 + 0.66667) = 0.375 \text{ A}$$

$$I_2 = -0.375(2/3) = -0.25 \text{ A.}$$

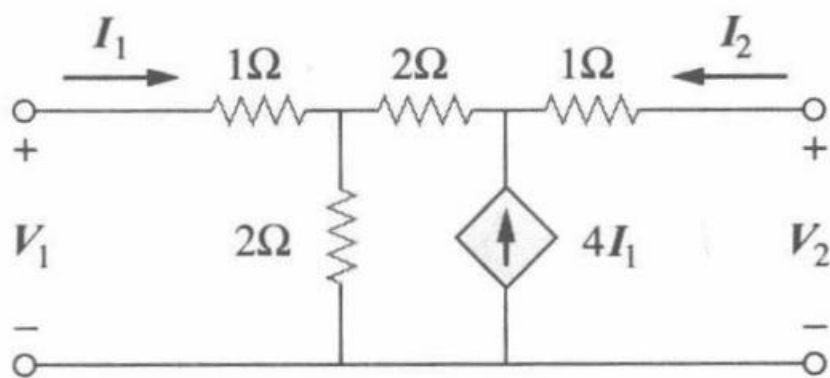
$$y_{11} = 0.375/1 = \mathbf{0.375 \text{ S}}, \quad y_{21} = \mathbf{-0.25 \text{ S.}}$$

$$I_2 = 1 / [(1/1) + (1/(0.5+0.5))] = 0.5 \text{ A}$$

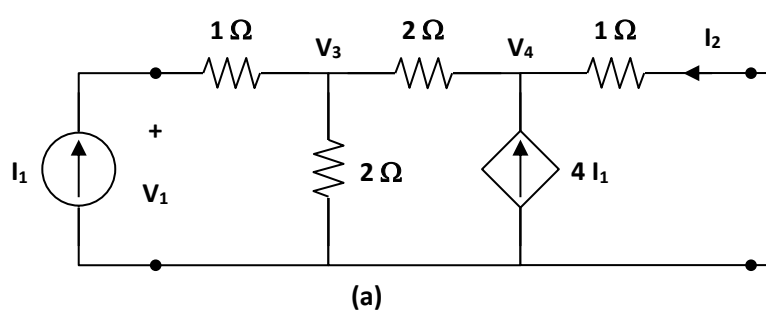
$$I_1 = -0.5(1/2) = \mathbf{-0.25 \text{ A.}}$$

$$y_{12} = \mathbf{-0.25 \text{ S}} \text{ and } y_{22} = \mathbf{0.5 \text{ S.}}$$

19.31 求下图所示电路的混合参数



解:



在节点 1:

$$I_1 = \frac{V_3}{2} + \frac{V_3 - V_4}{2} \longrightarrow 2I_1 = 2V_3 - V_4$$

在节点 2:

$$\frac{V_3 - V_4}{2} + 4I_1 = \frac{V_4}{1}$$

$$8I_1 = -V_3 + 3V_4 \longrightarrow 16I_1 = -2V_3 + 6V_4$$

联立:

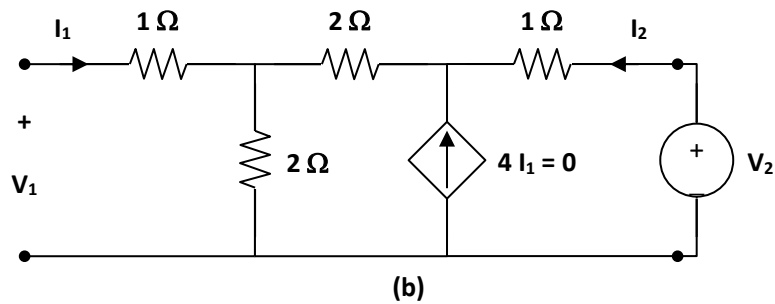
$$18I_1 = 5V_4 \longrightarrow V_4 = 3.6I_1$$

$$V_3 = 3V_4 - 8I_1 = 2.8I_1$$

$$V_1 = V_3 + I_1 = 3.8I_1$$

$$h_{11} = \frac{V_1}{I_1} = 3.8 \Omega$$

$$I_2 = \frac{-V_4}{1} = -3.6I_1 \longrightarrow h_{21} = \frac{I_2}{I_1} = -3.6$$

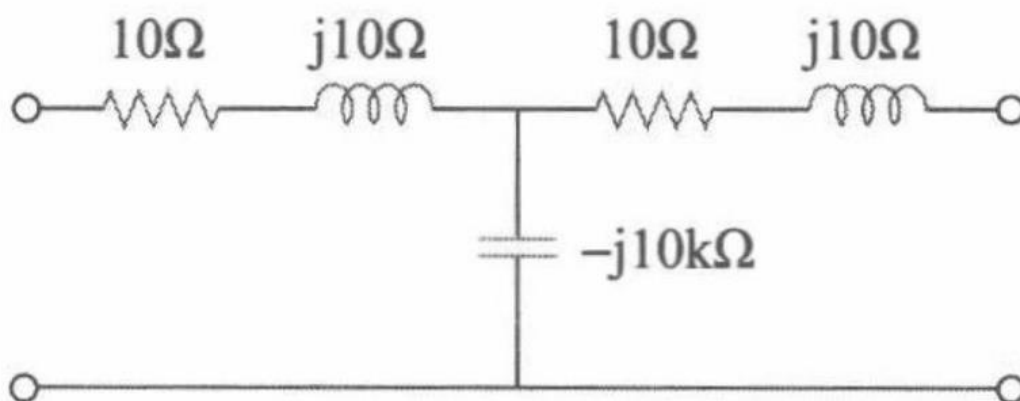


$$V_1 = \frac{2}{2+2+1} V_2 = \frac{2}{5} V_2 \longrightarrow h_{12} = \frac{V_1}{V_2} = 0.4$$

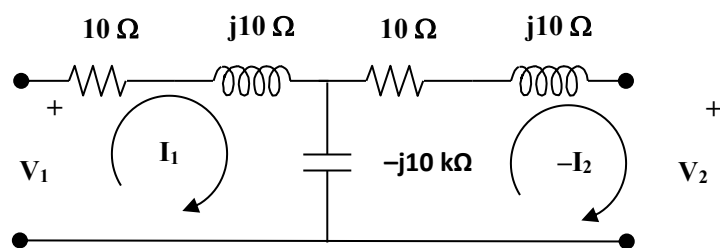
$$I_2 = \frac{V_2}{2+2+1} = \frac{V_2}{5} \longrightarrow h_{22} = \frac{I_2}{V_2} = \frac{1}{5} = 0.2 \text{ S}$$

$$[h] = \begin{bmatrix} 38 \Omega & 0.4 \\ -3.6 & 0.2 \text{ S} \end{bmatrix}$$

19.45 求下图所示电路的 ABCD 参数



解:



$$V_1 = AV_2 - BI_2, \quad I_1 = CV_2 - DI_2.$$

$$\text{令 } I_1 = 1 \text{ A}, \quad I_2 = 0, \quad I_1 = 1 \text{ A}, \quad V_2 = 0.$$

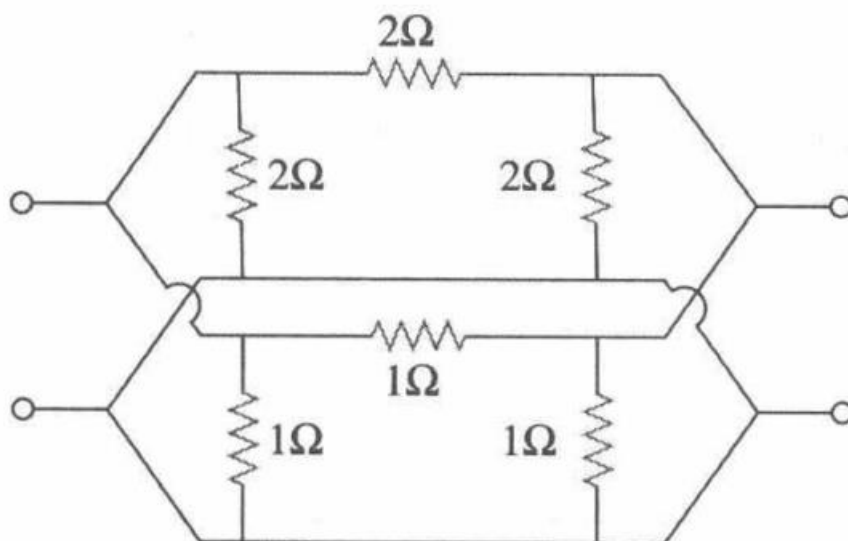
$$V_1 = (10+j10)1 - j10,000 \times 1 \approx -j10 \text{ kV and } V_2 = -j10 \text{ kV}$$

$$A = 1, \quad C = 1/(-j10\text{k}) = j100 \mu\text{S}.$$

$$V_1 \approx (10+j10+10+j10)1 = (20+j20) \text{ V}, \quad I_2 \approx -1 \text{ A which leads to}$$

$$B = (20+j20) \Omega, \quad D = 1.$$

19.68 求下图所示网络的 h 参数



解:

$$\text{对于上面的网络 } N_a, \quad [y_a] = \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}$$

$$\text{对于下面的网络 } N_b, \quad [y_b] = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$$

对于总网络,

$$[y] = [y_a] + [y_b] = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$\Delta_y = 36 - 9 = 27$$

$$[\mathbf{h}] = \begin{bmatrix} \frac{1}{\mathbf{y}_{11}} & \frac{-\mathbf{y}_{12}}{\mathbf{y}_{11}} \\ \frac{\mathbf{y}_{21}}{\mathbf{y}_{11}} & \frac{\Delta_y}{\mathbf{y}_{11}} \end{bmatrix} = \begin{bmatrix} \frac{1}{6}\Omega & \frac{1}{2} \\ \frac{1}{2} & \frac{9}{2}\mathbf{S} \end{bmatrix}$$